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(54) **WEARABLE DEVICE FOR COUPLING TO A USER, AND MEASURING AND MONITORING USER ACTIVITY**

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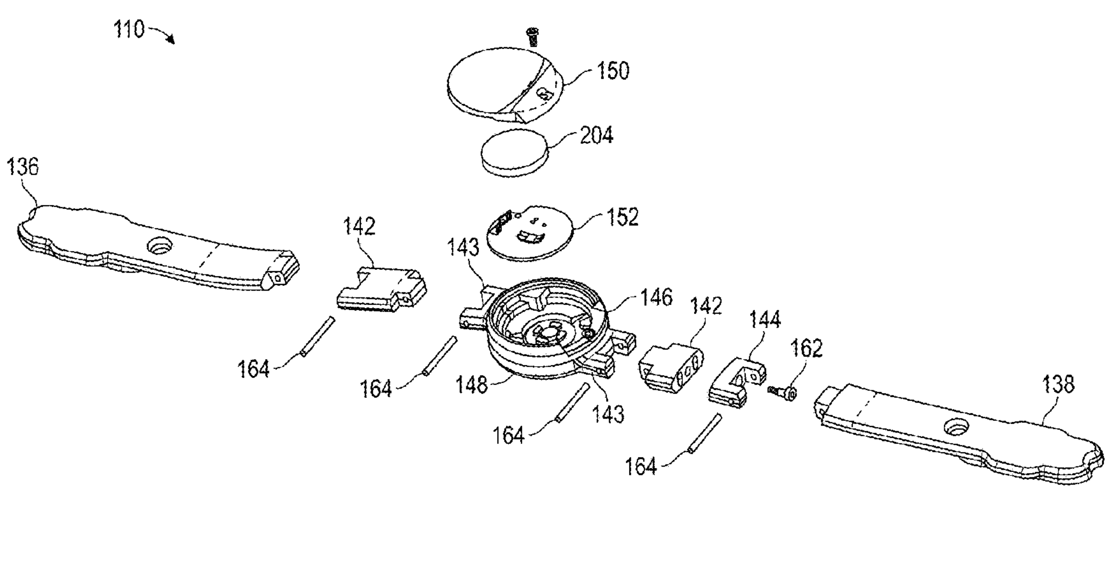
(57) **ABSTRACT**

A system for measuring an angle of a joint of a user includes a center hub, a first arm, a second arm, a magnet, and a sensor. The center hub includes a first hub and a second hub. The first arm is configured for attachment to a first limb portion of the user at a first outer end and to the first hub at a first inner end. The second arm is configured for attachment to a second limb portion of the user at a second outer end and to the second hub at a second inner end, wherein the first hub is pivotally coupled to the second hub. The magnet is coupled to the second hub. The sensor is disposed in the center hub and configured to detect a rotation of the magnet.

20 Claims, 29 Drawing Sheets



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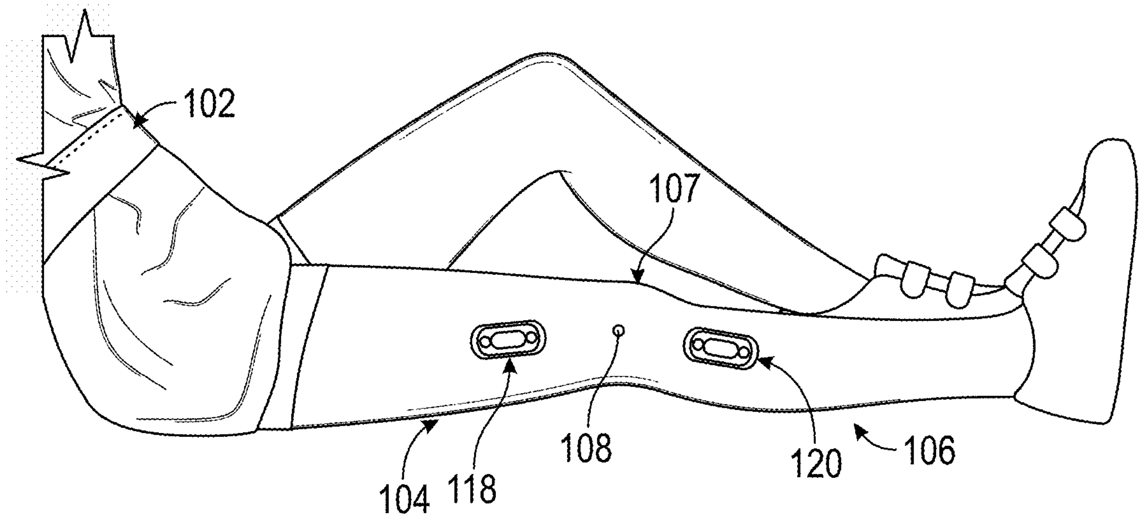

FIG. 1A

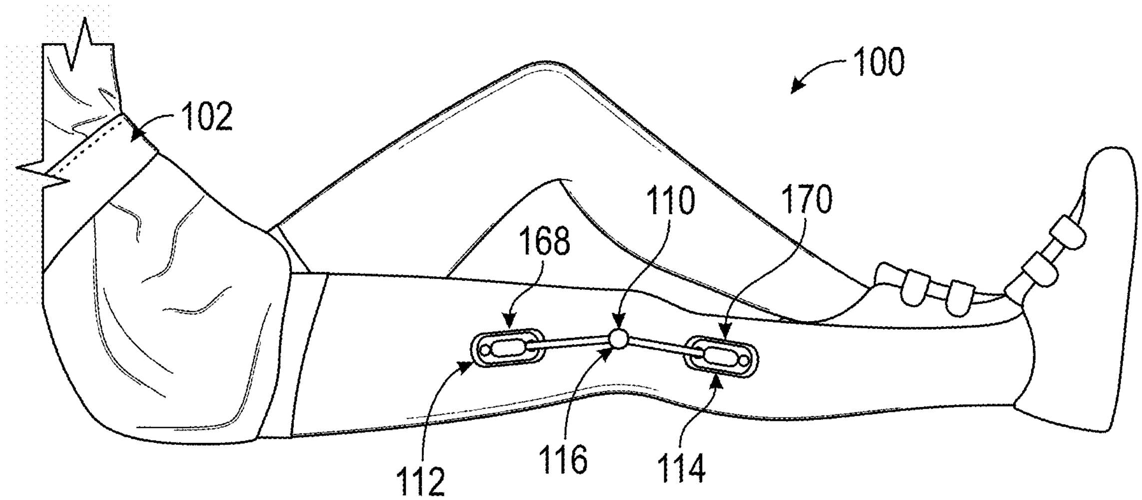

FIG. 1B

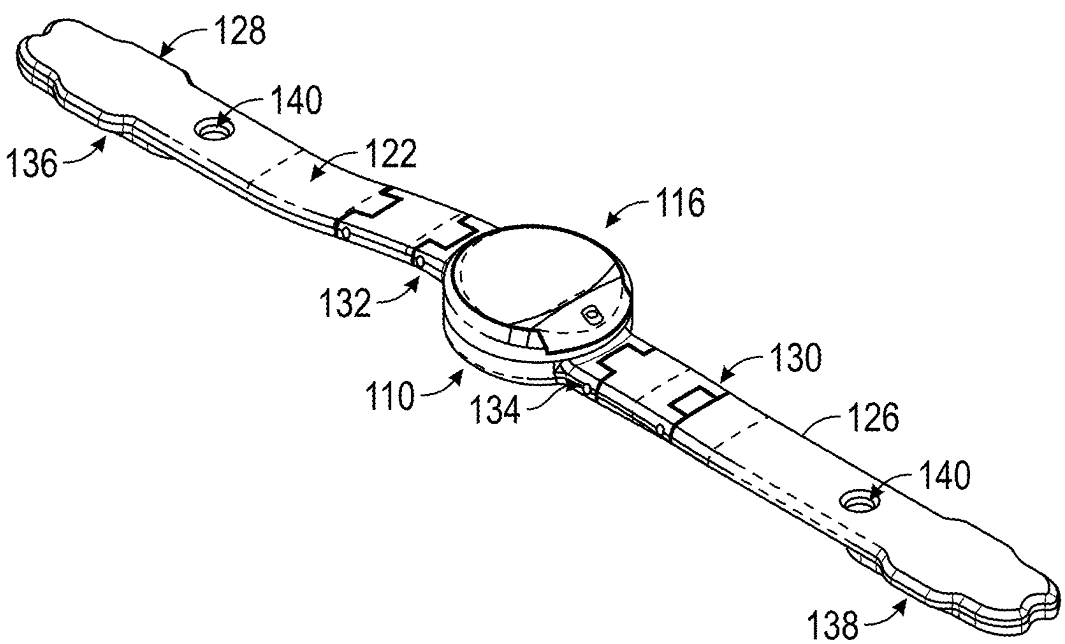
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FIG. 2A

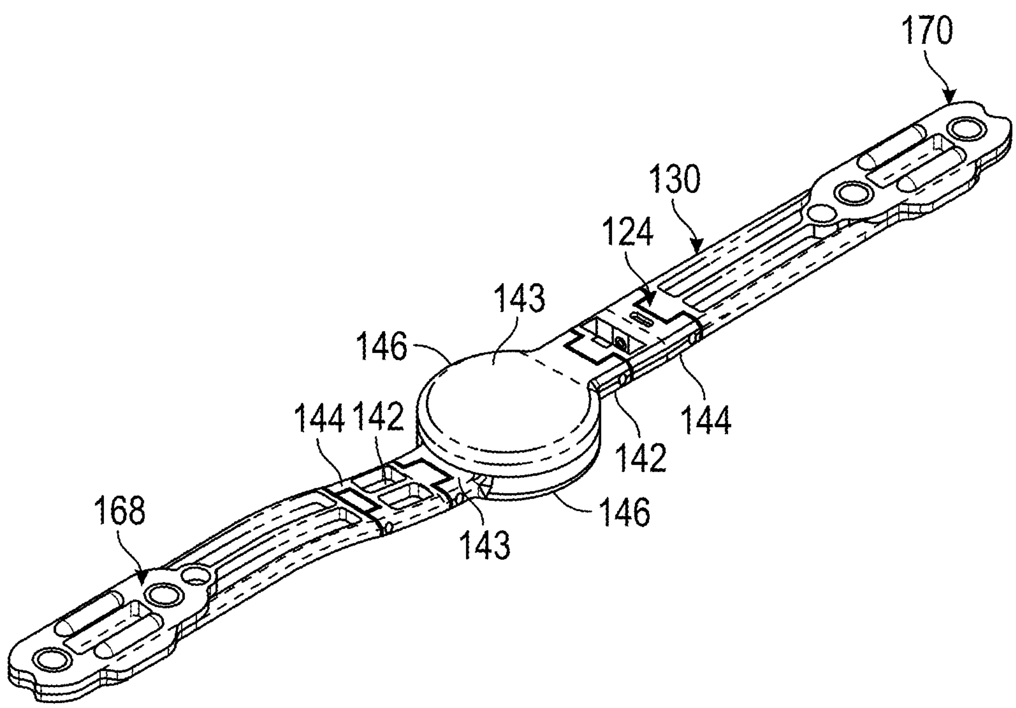
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FIG. 2B

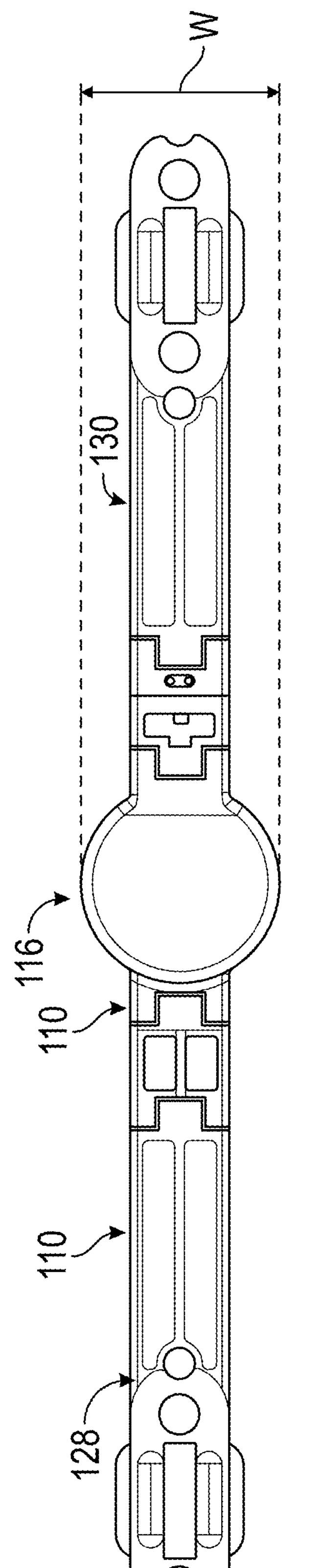

FIG. 3A

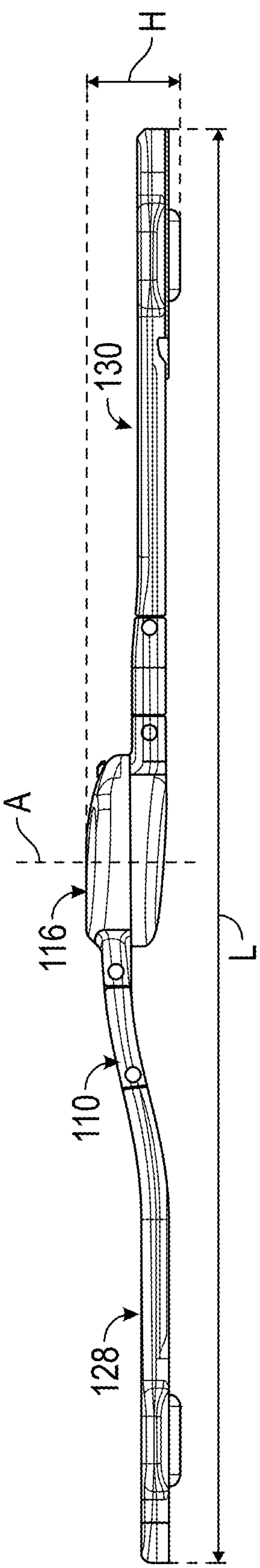

FIG. 3B

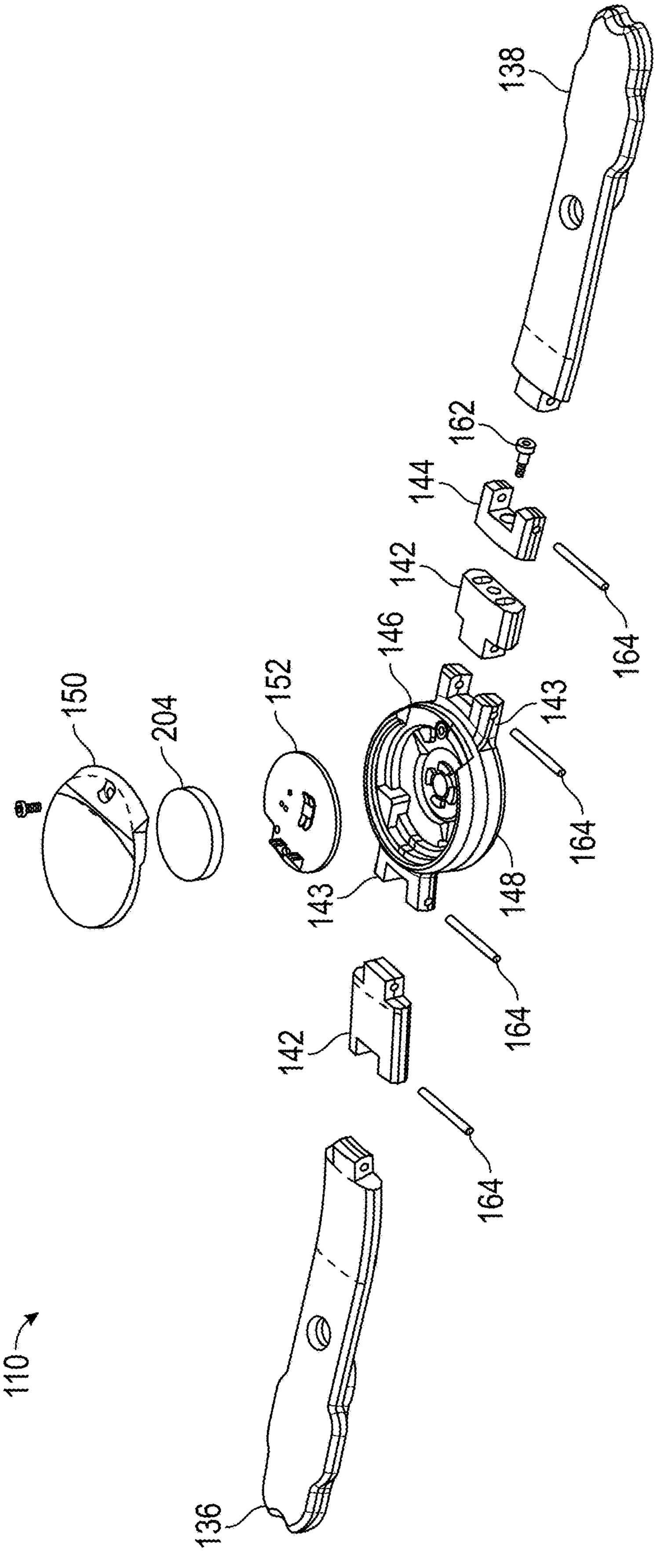

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FIG. 4

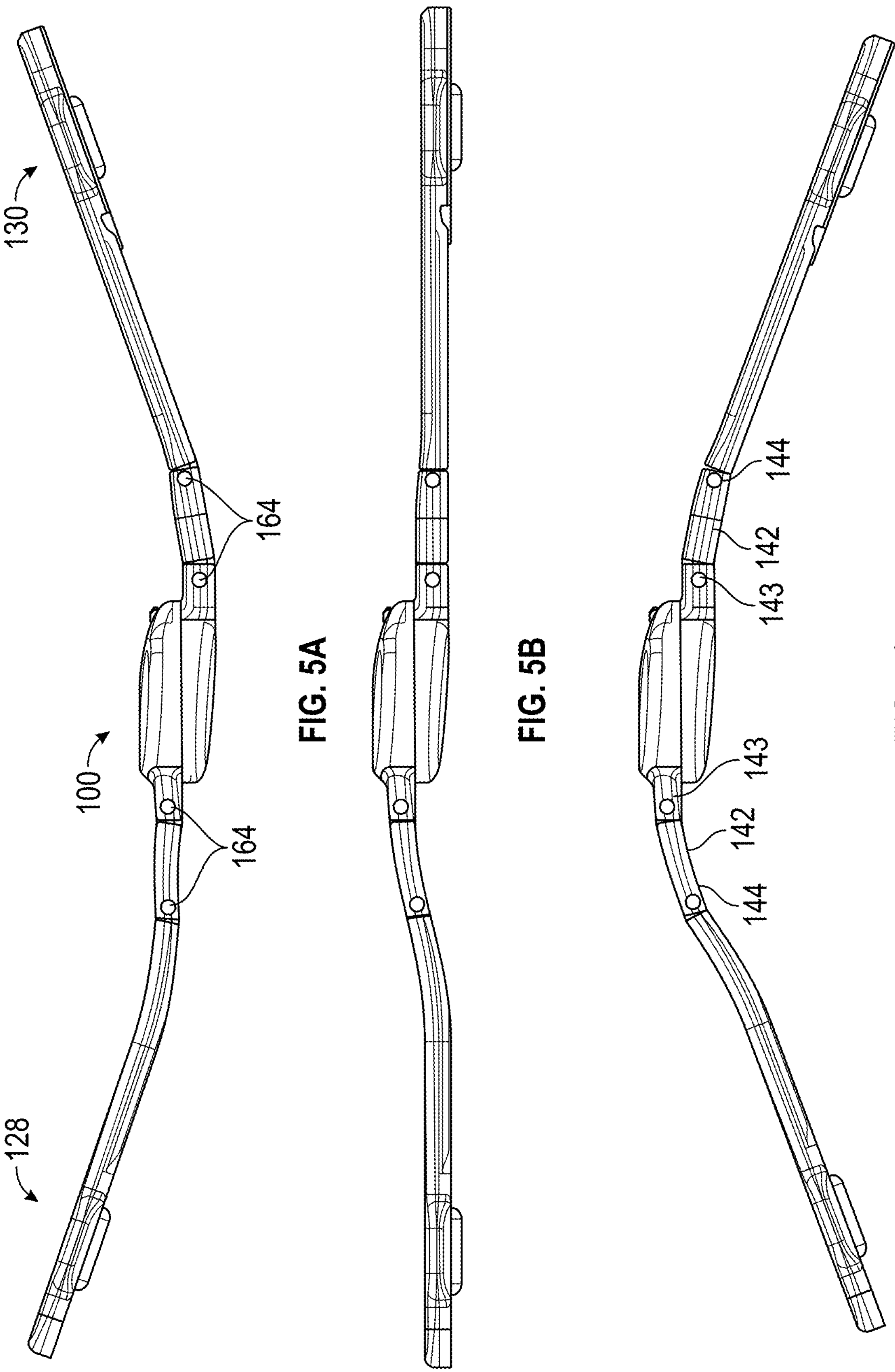


FIG. 5A

FIG. 5B

FIG. 5C

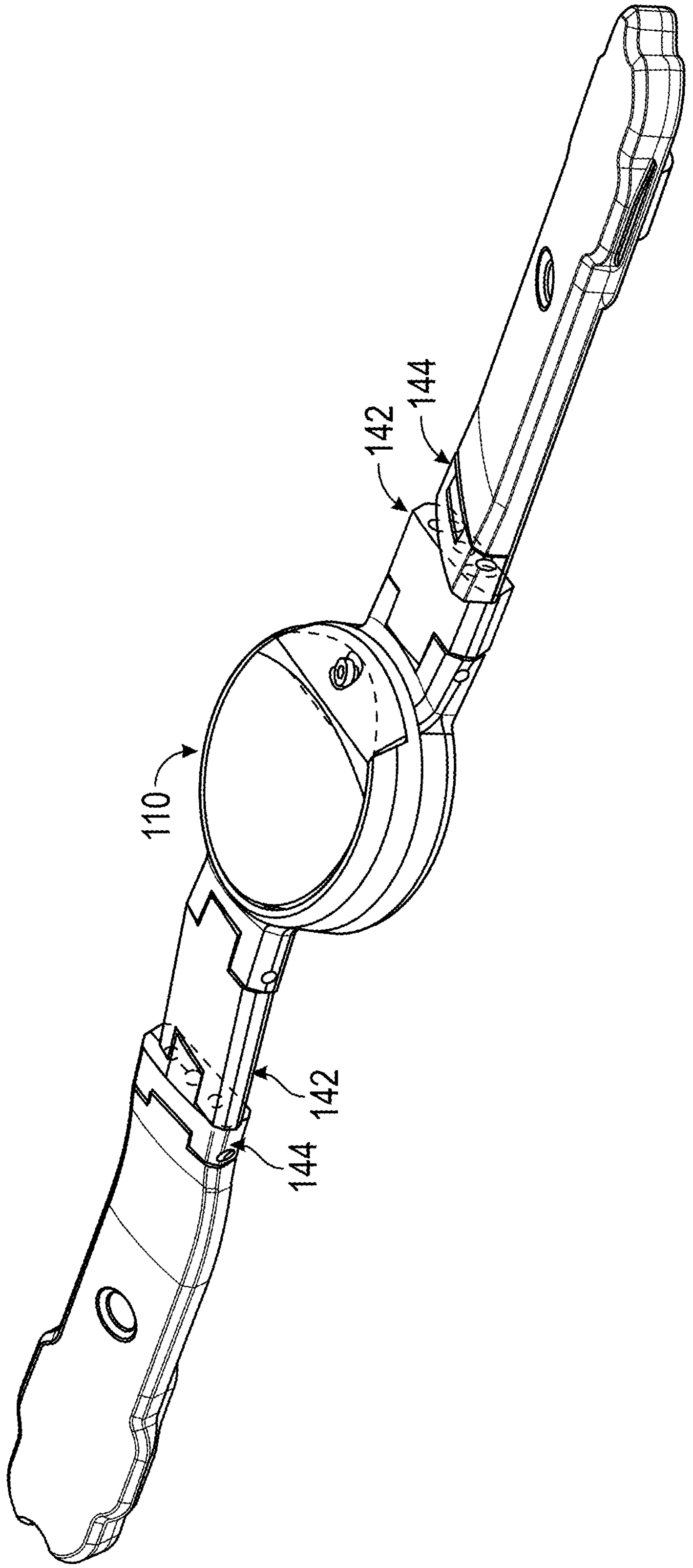


FIG. 6

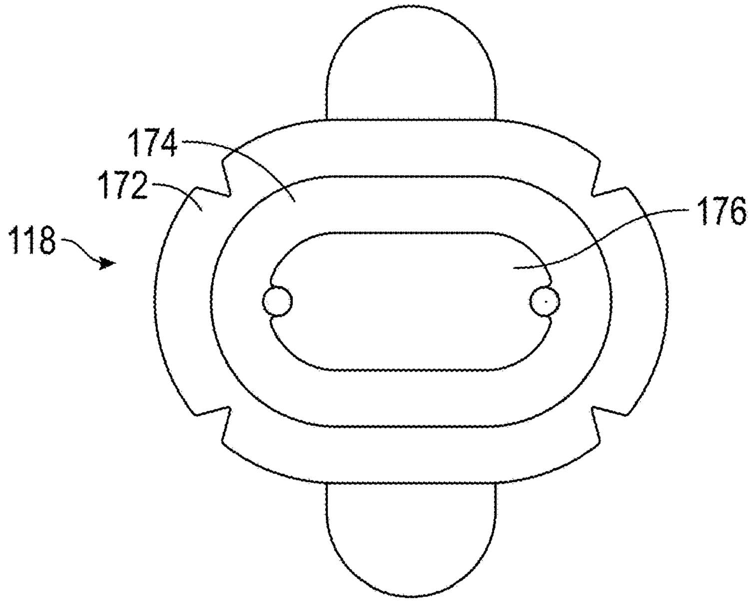


FIG. 7A

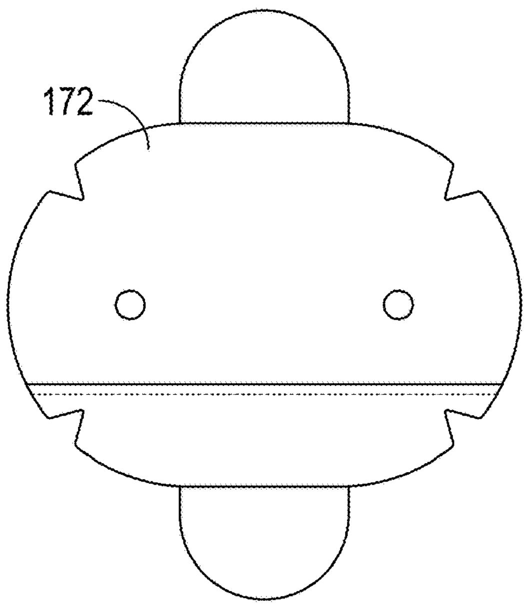


FIG. 7B

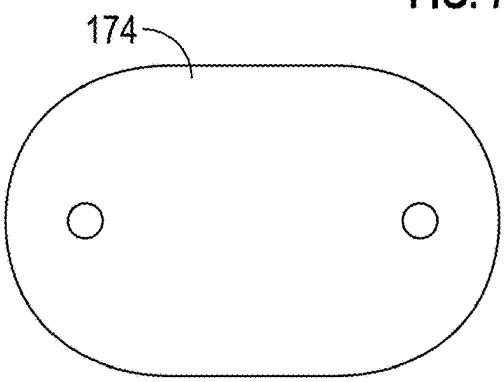


FIG. 7C

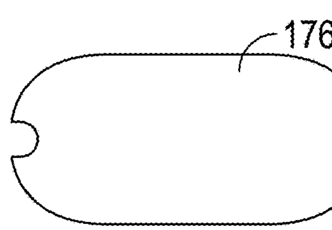


FIG. 7D

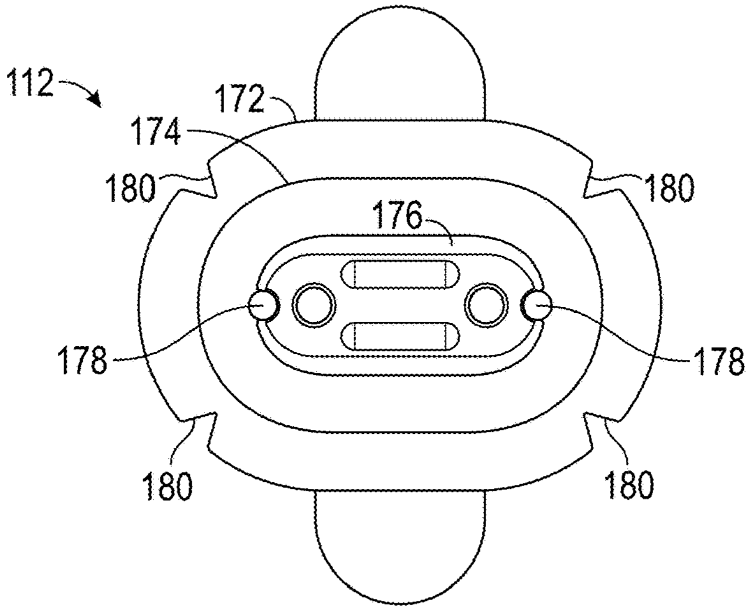
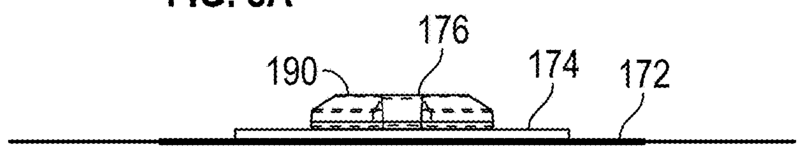


FIG. 8A



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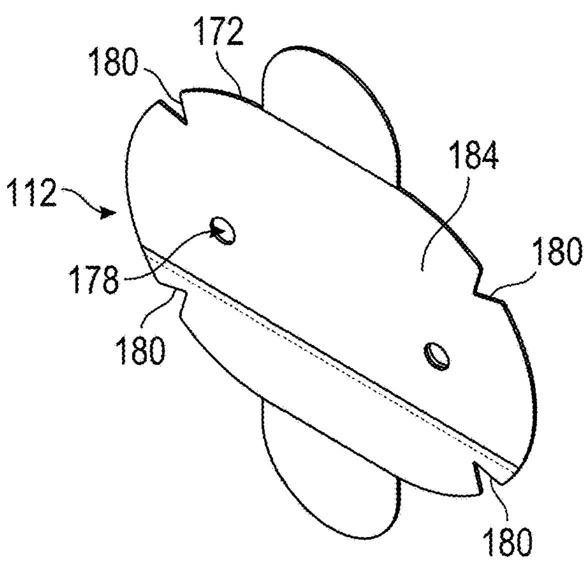


FIG. 8C

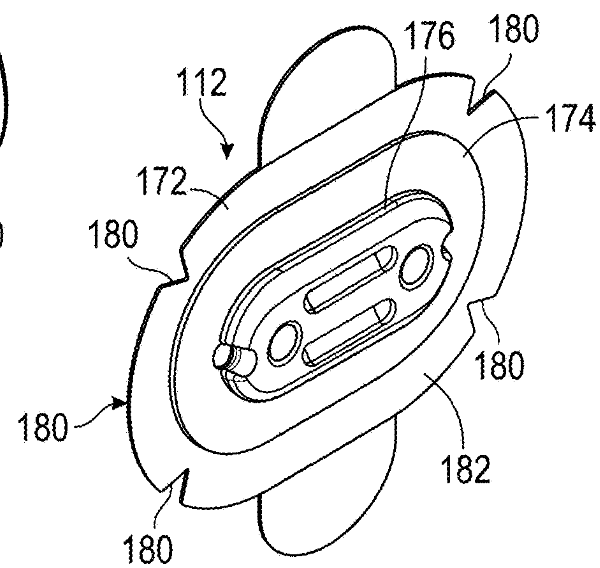


FIG. 8D

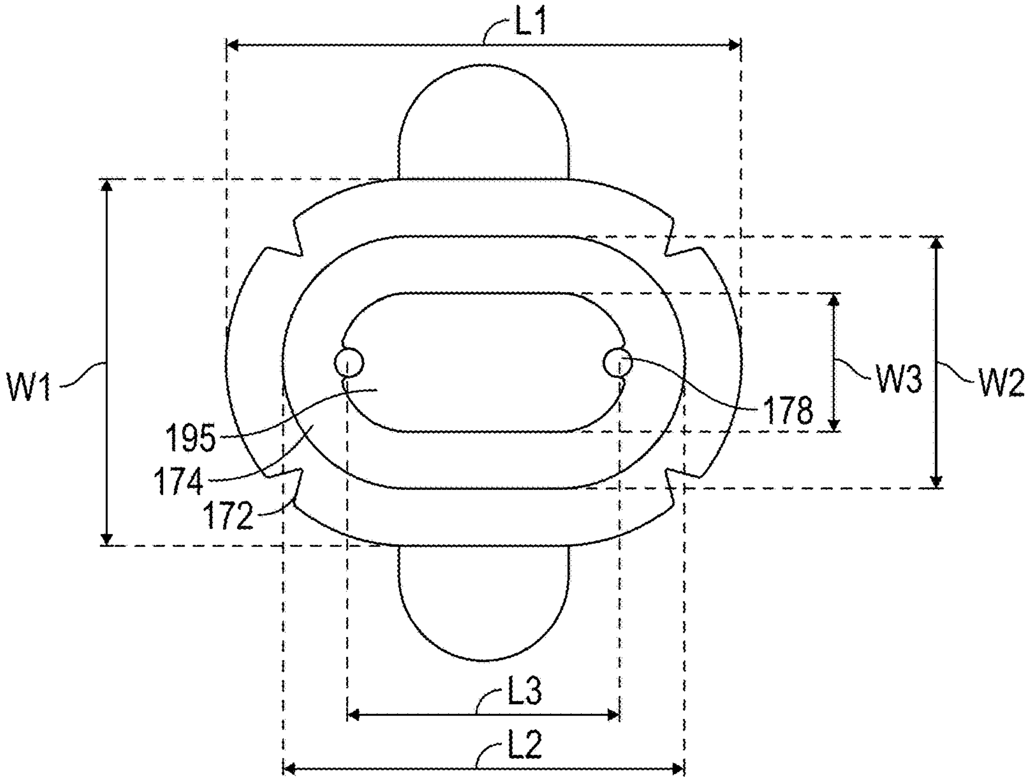


FIG. 9A

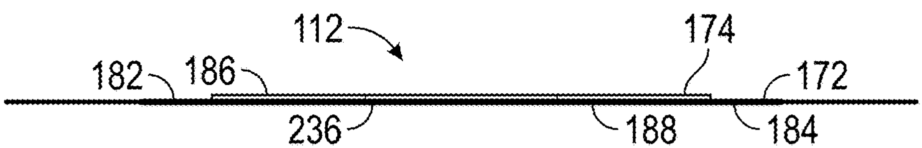


FIG. 9B

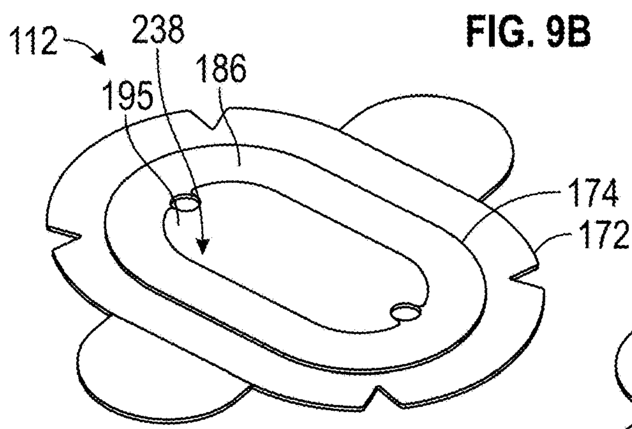


FIG. 9C

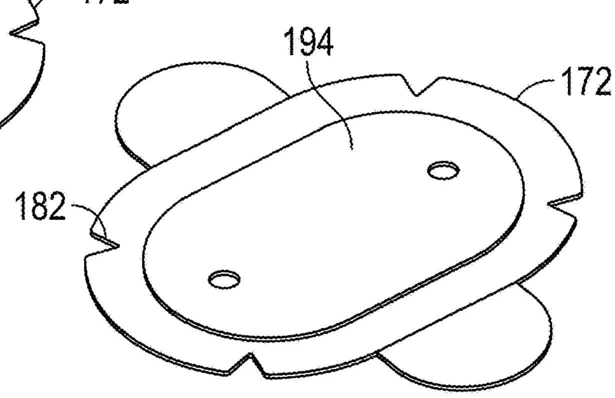


FIG. 9D

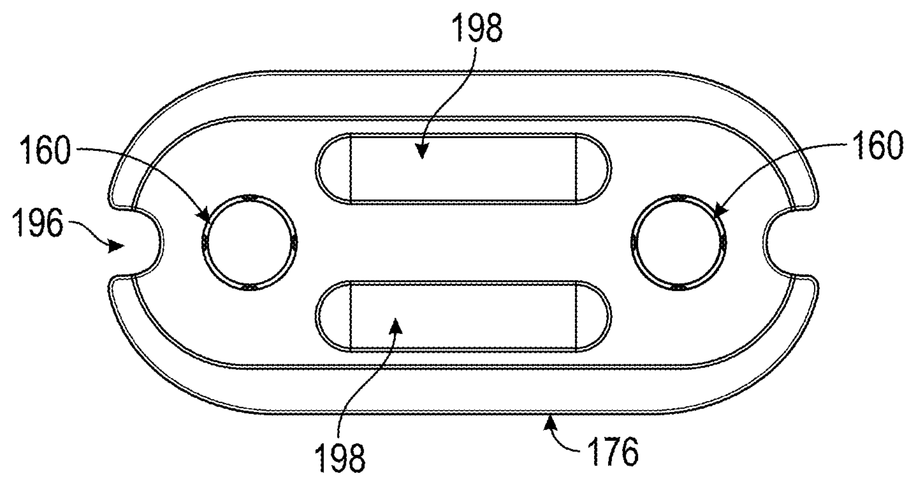


FIG. 10A

FIG. 10B

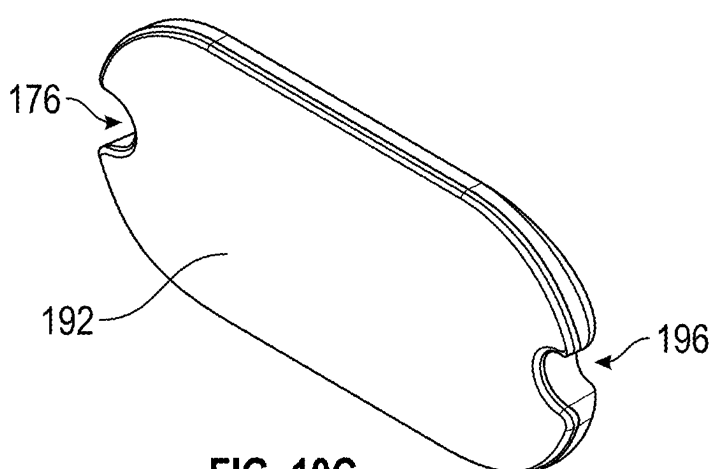


FIG. 10C

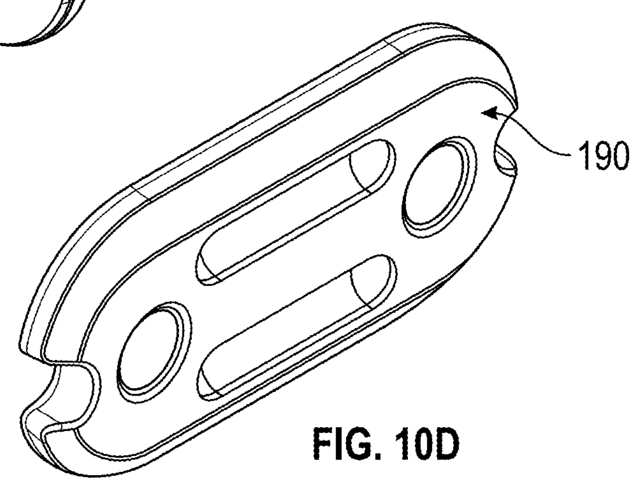


FIG. 10D

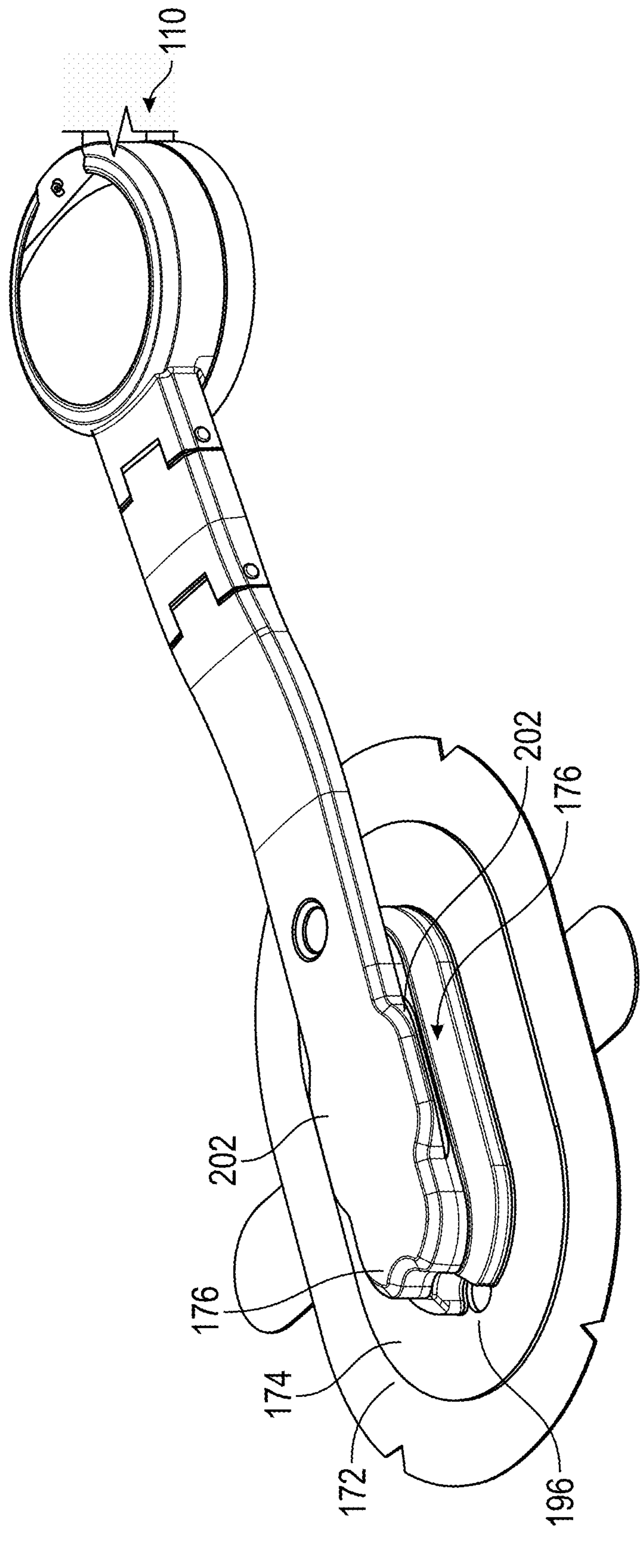
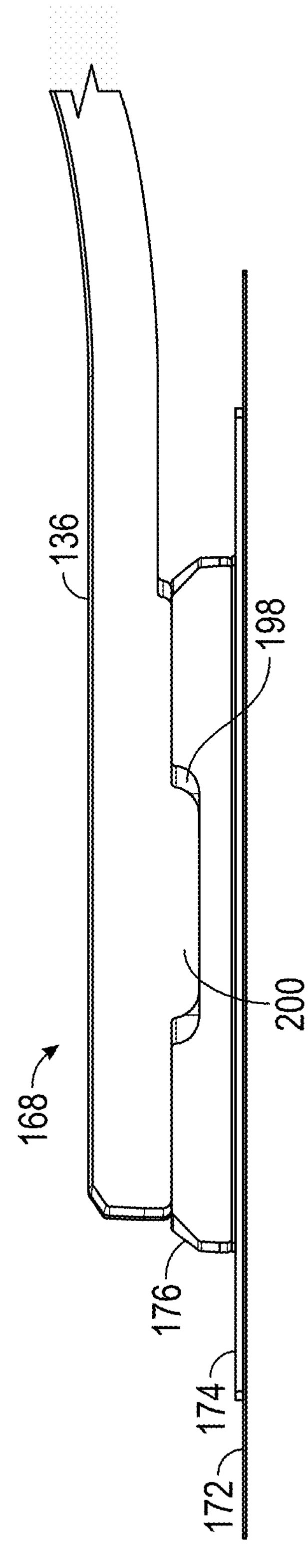


FIG. 11A



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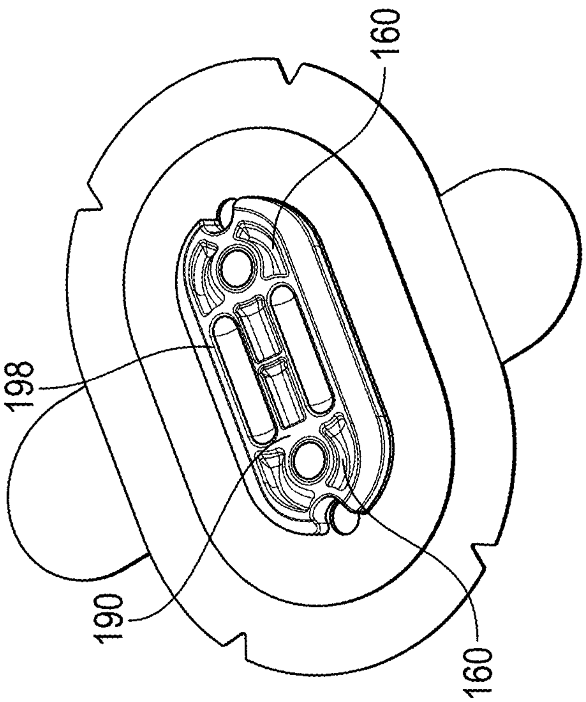
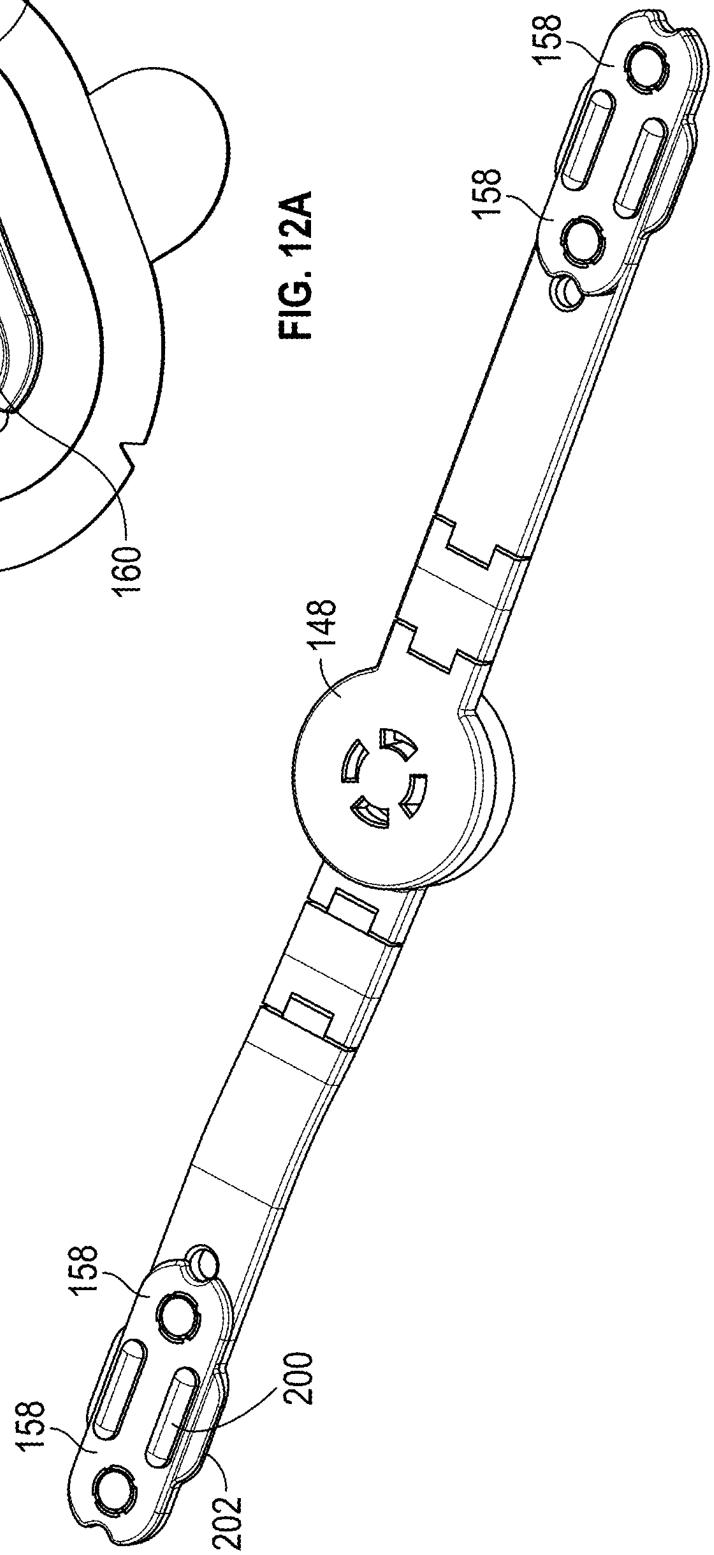


FIG. 12A



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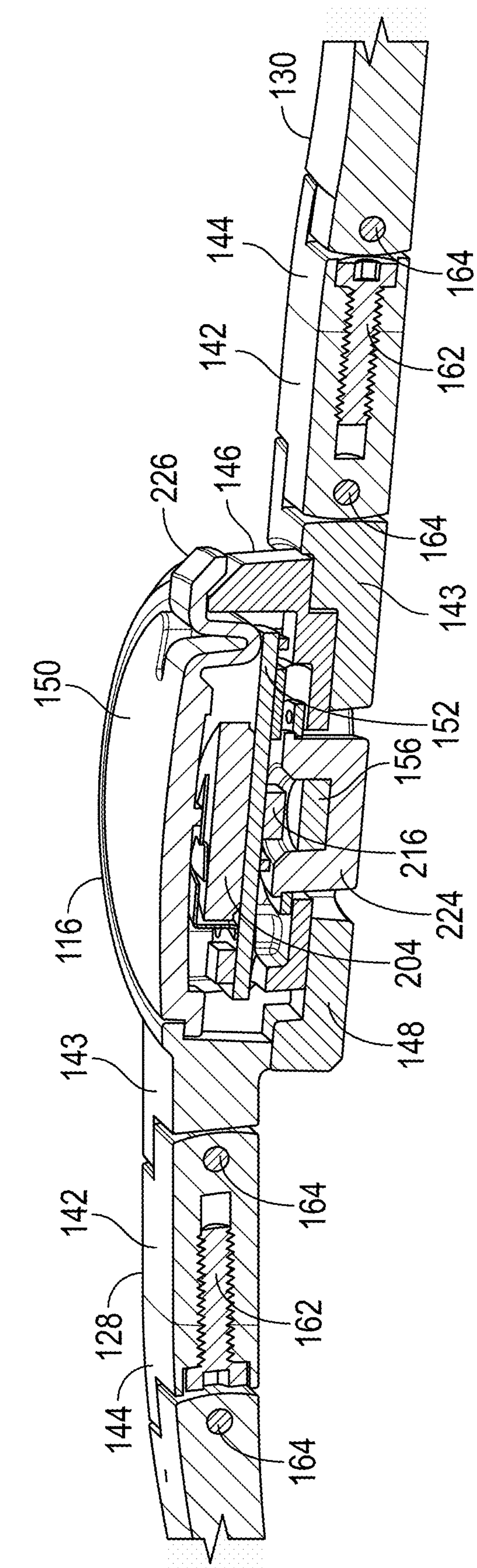
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FIG. 13

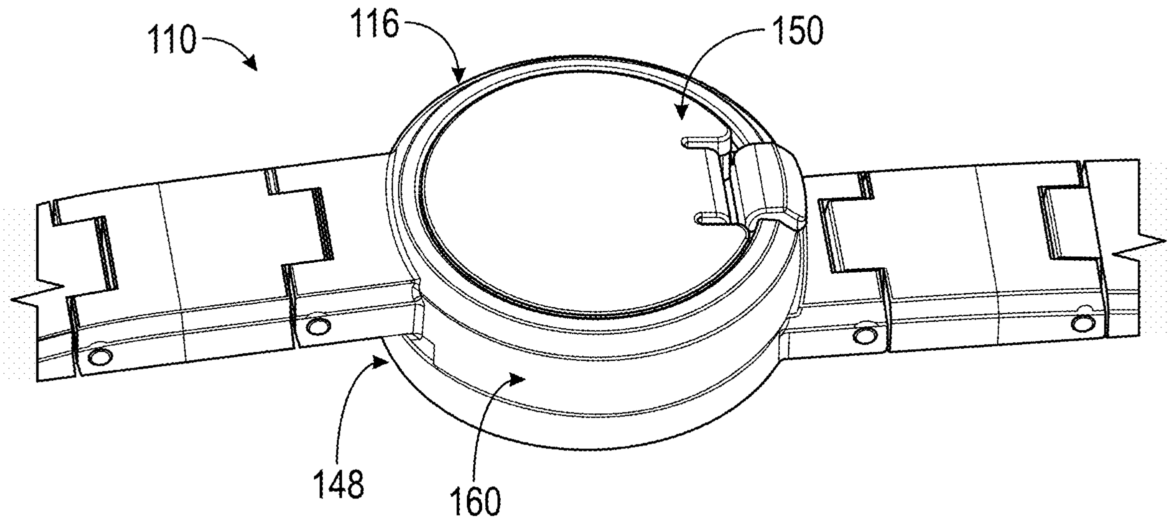
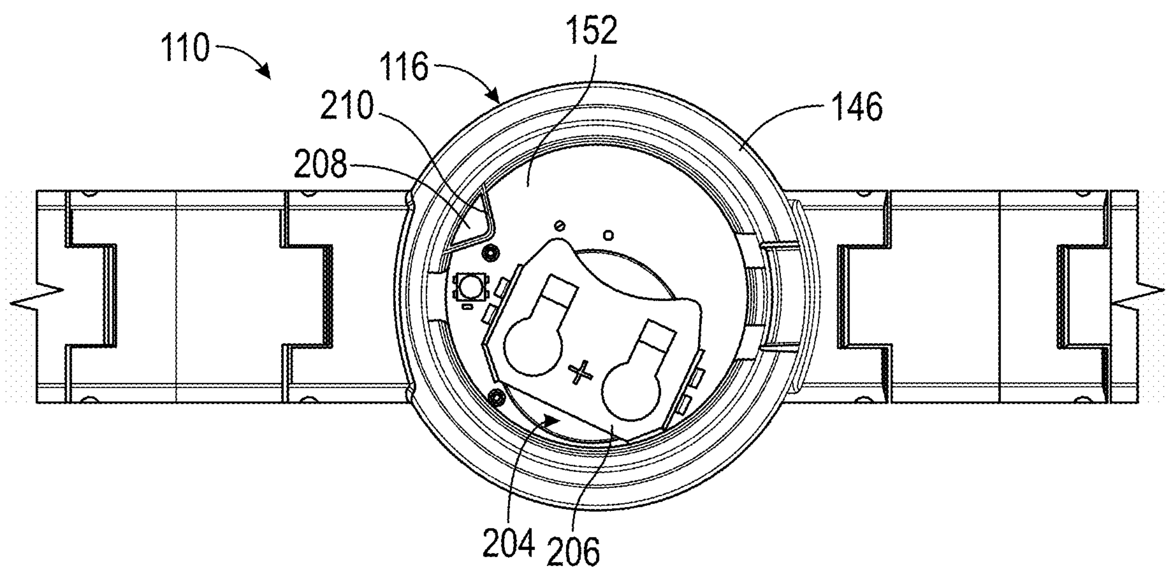


FIG. 14A



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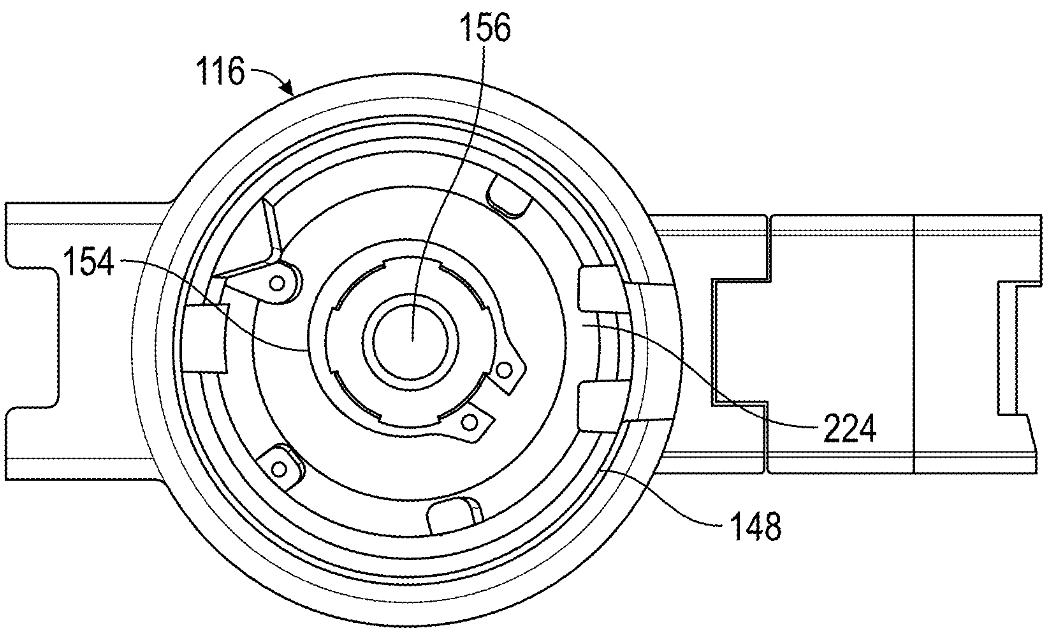
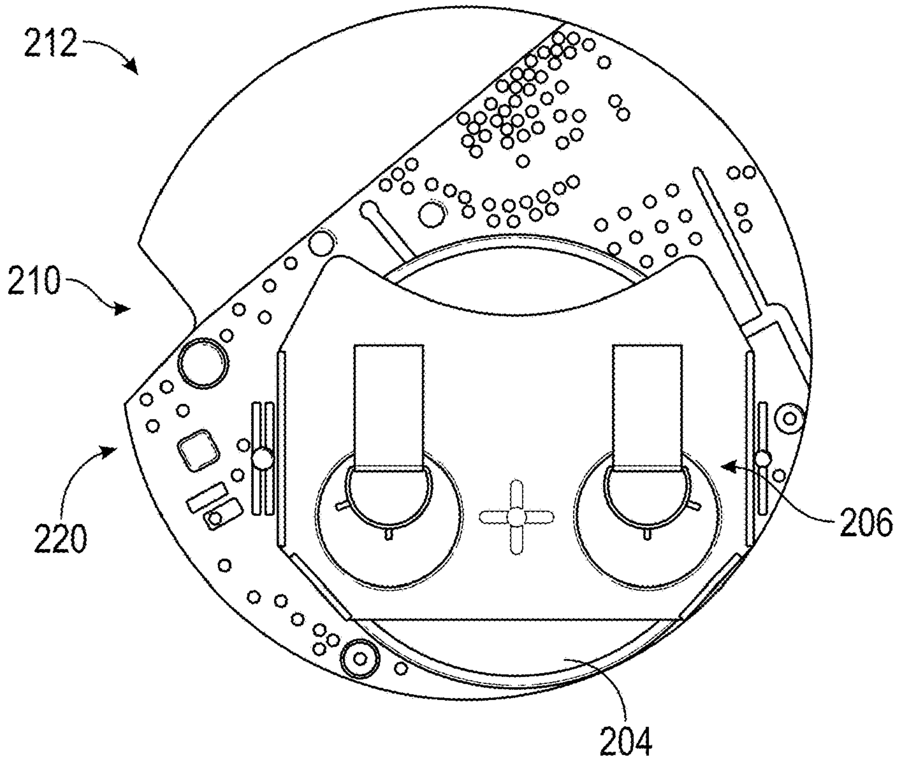


FIG. 14C



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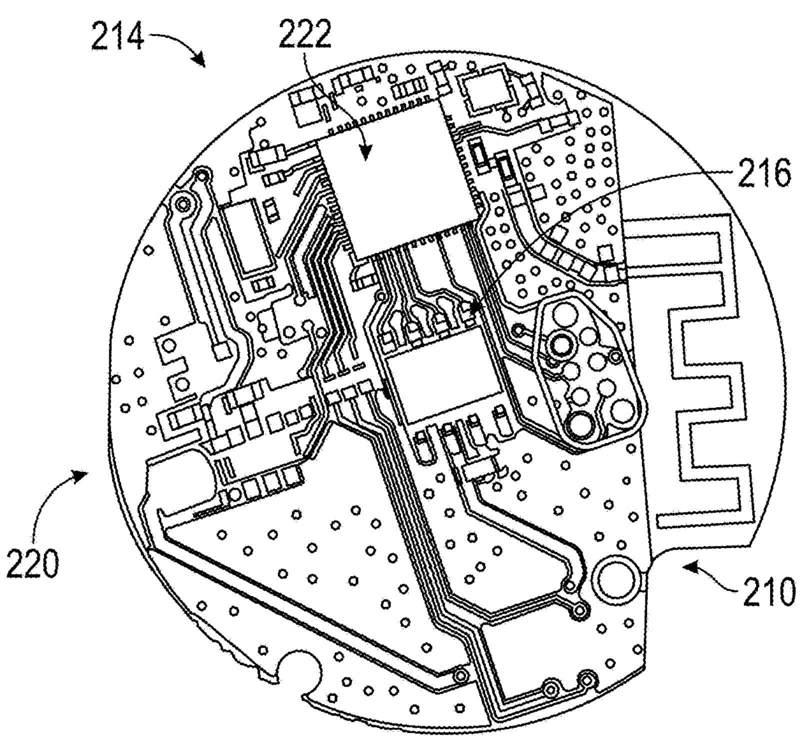

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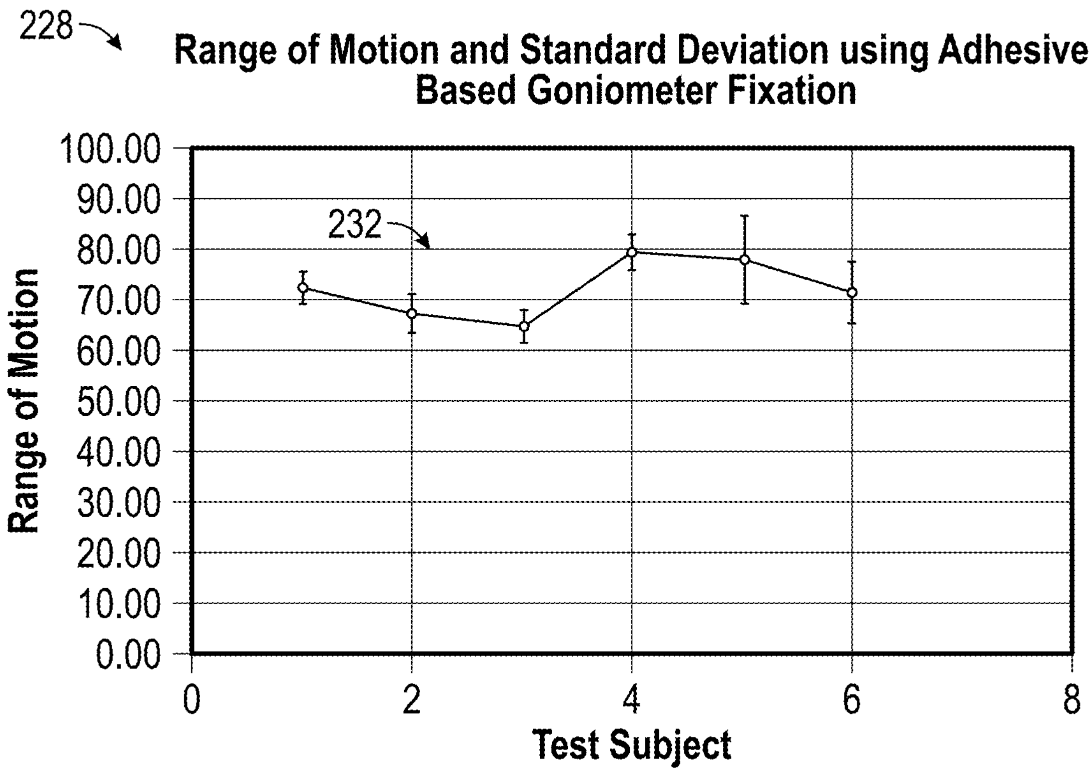

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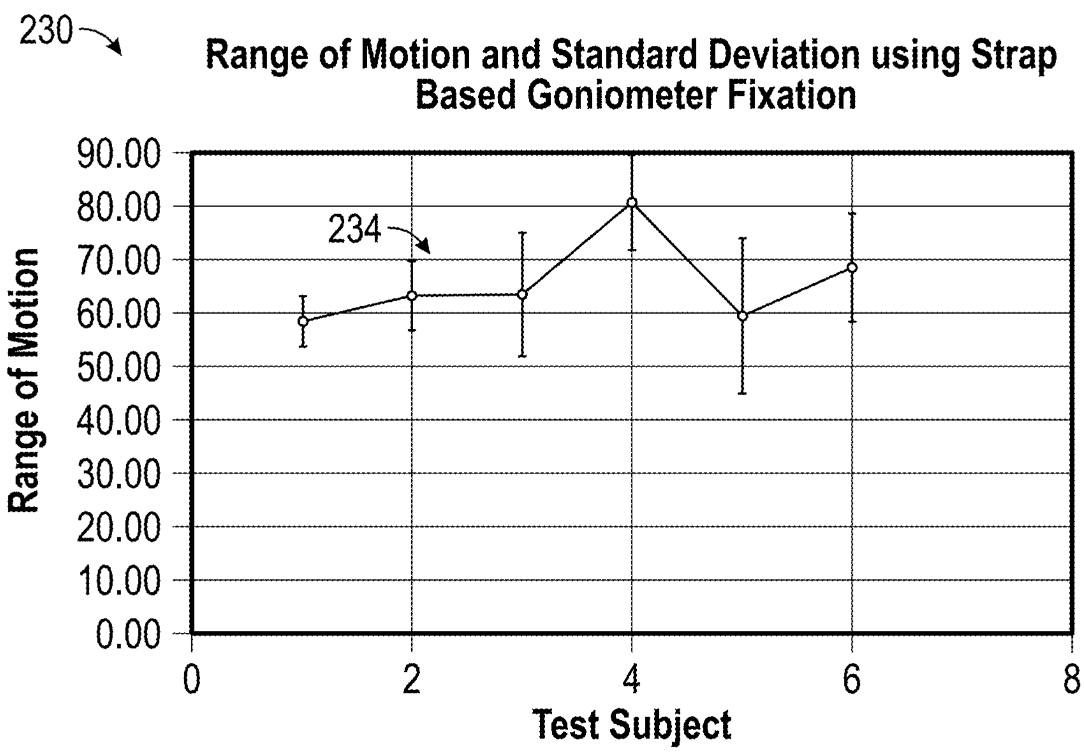

FIG. 16B

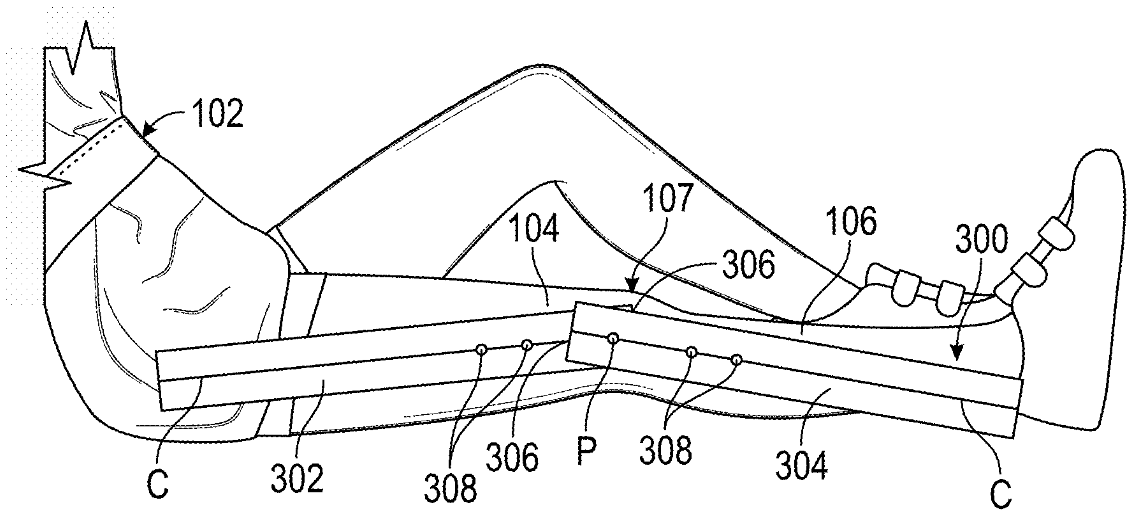
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FIG. 17

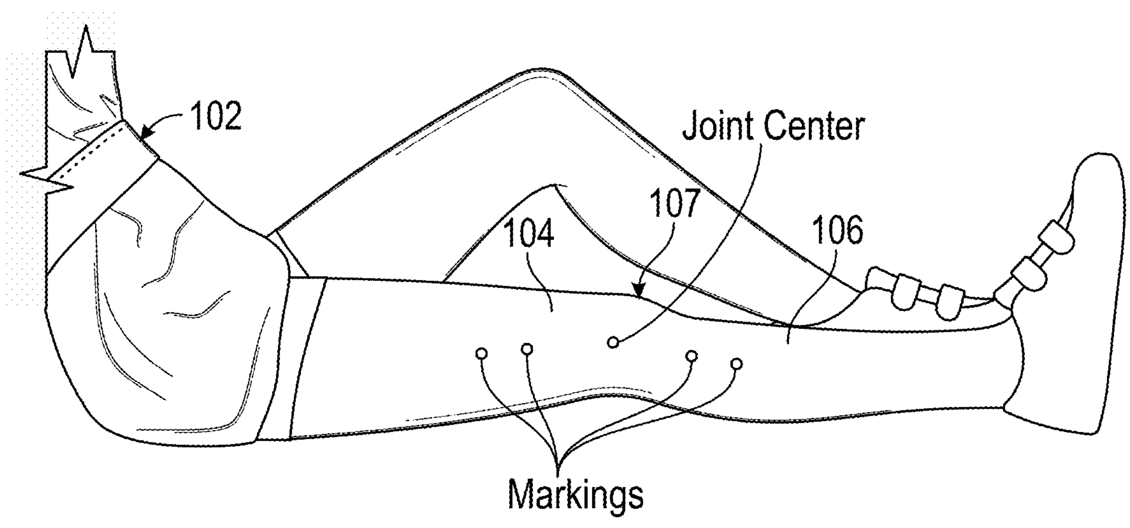
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FIG. 18

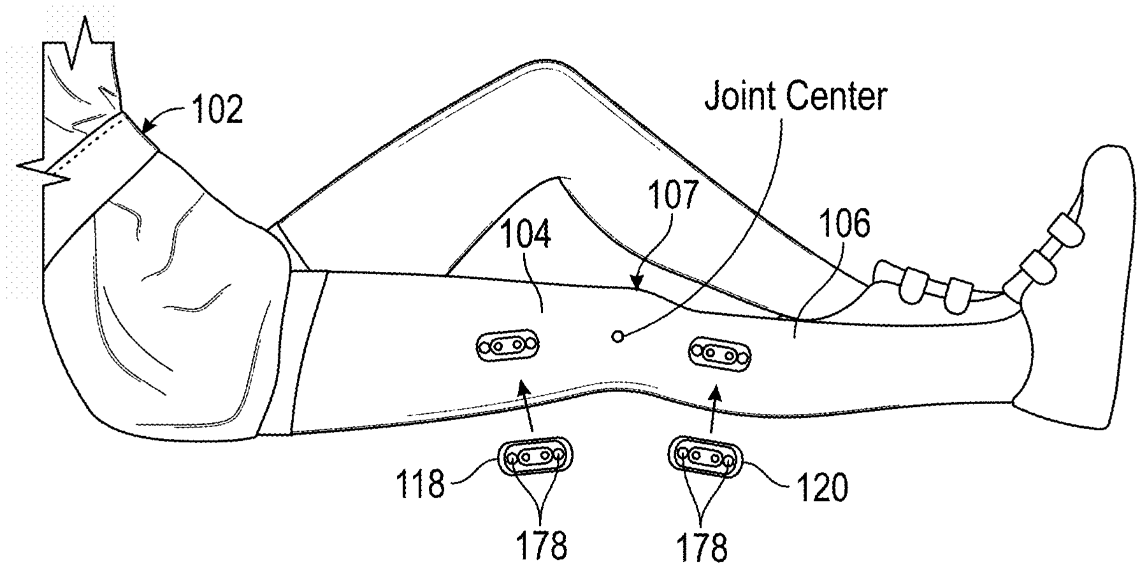
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FIG. 19

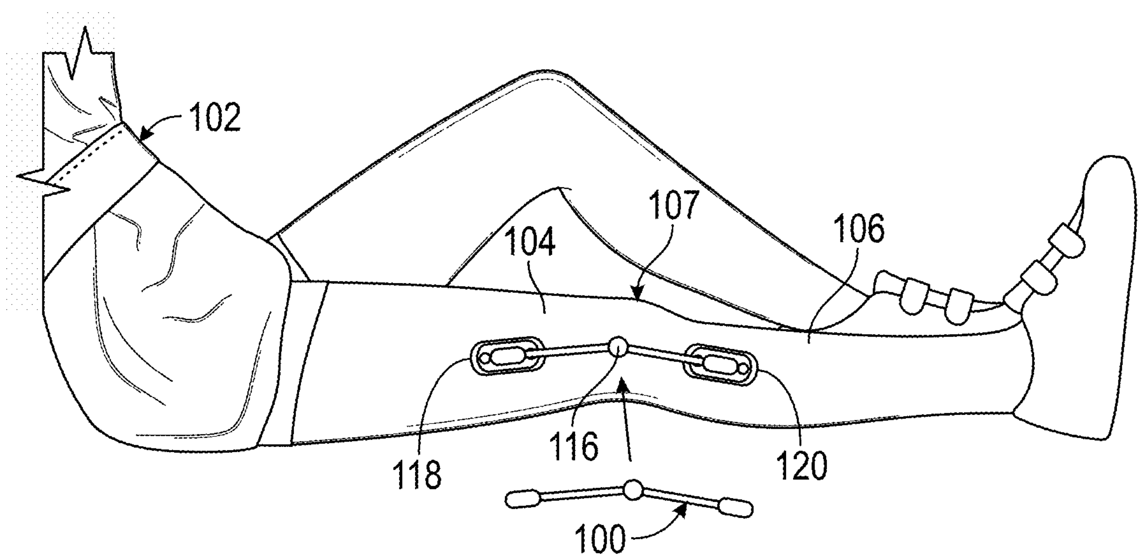
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FIG. 20

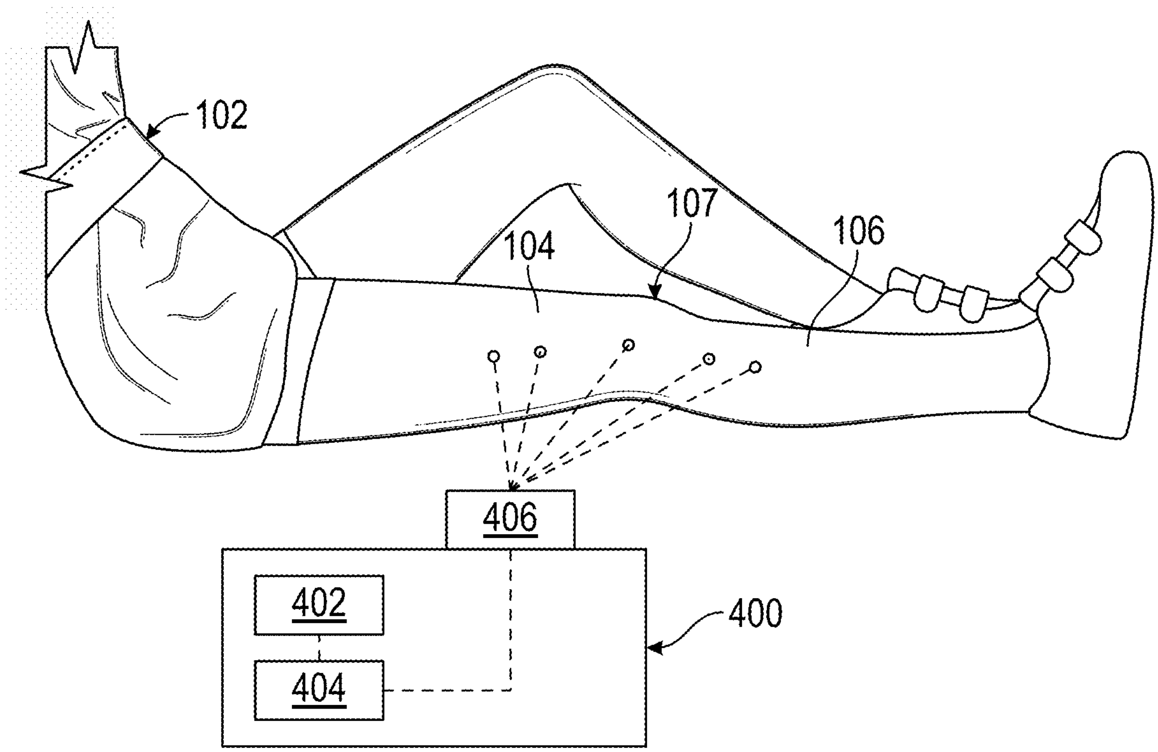
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FIG. 21

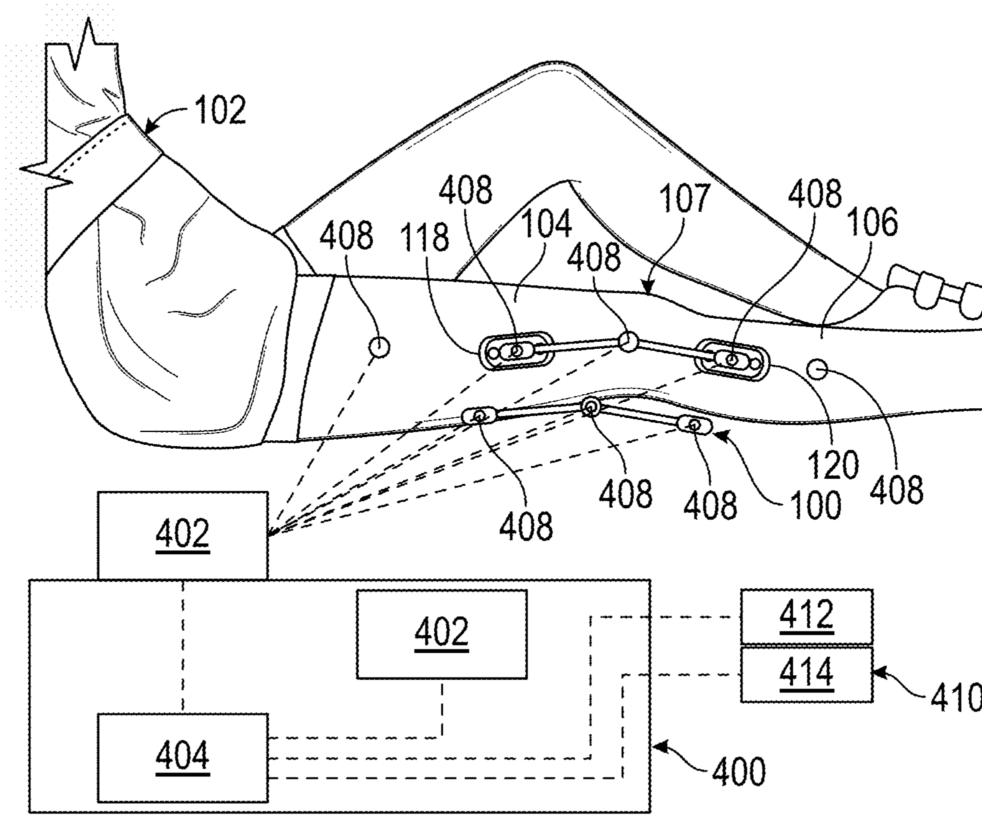
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FIG. 22

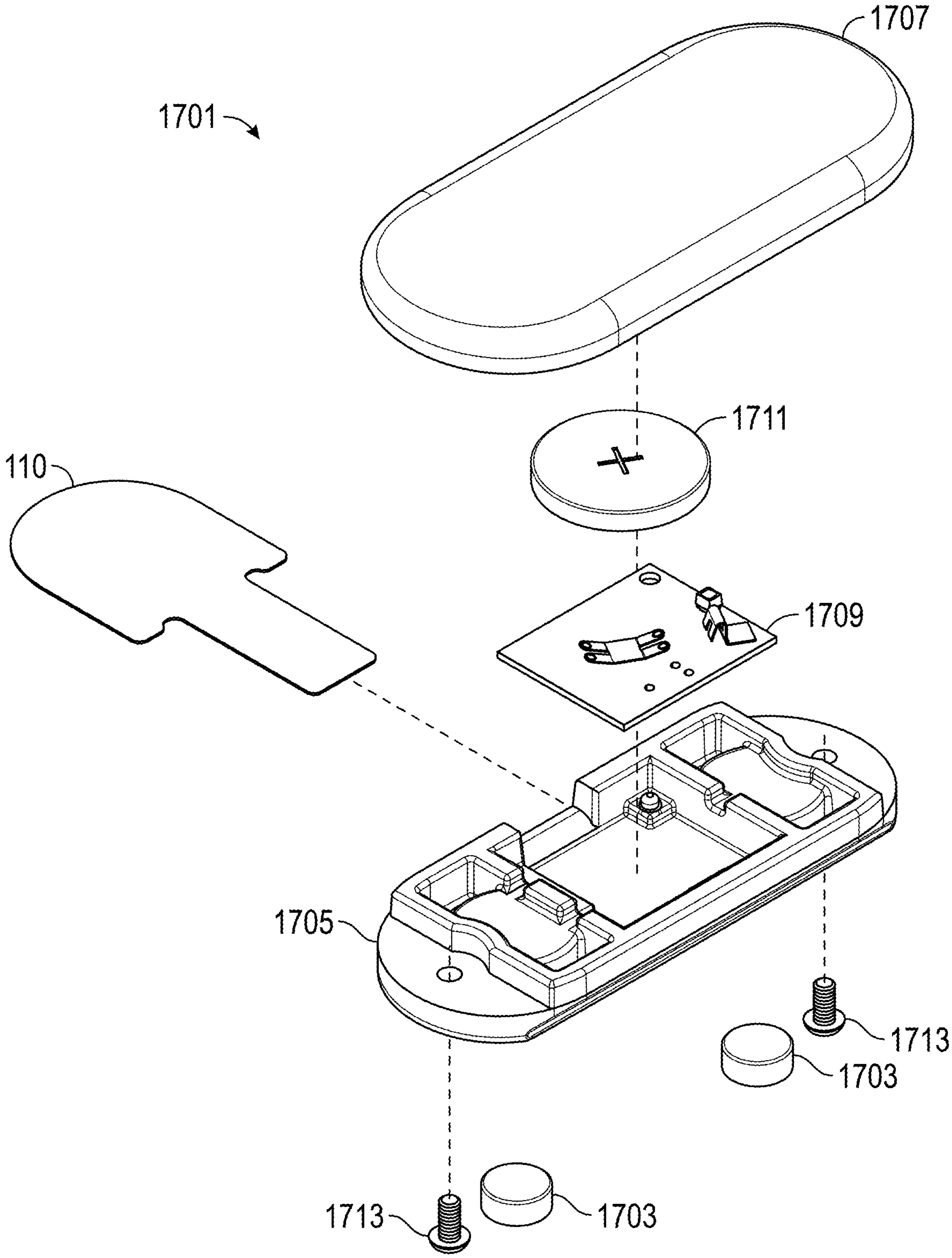

FIG. 23A

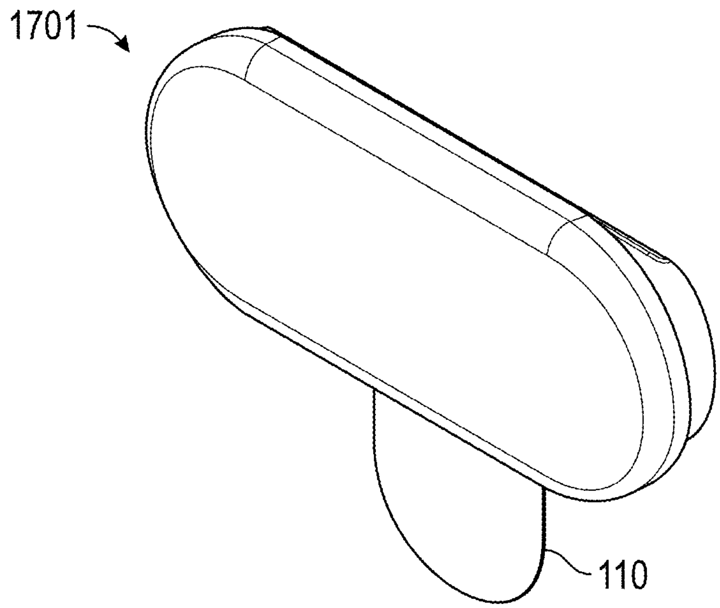

FIG. 23B

FIG. 23C

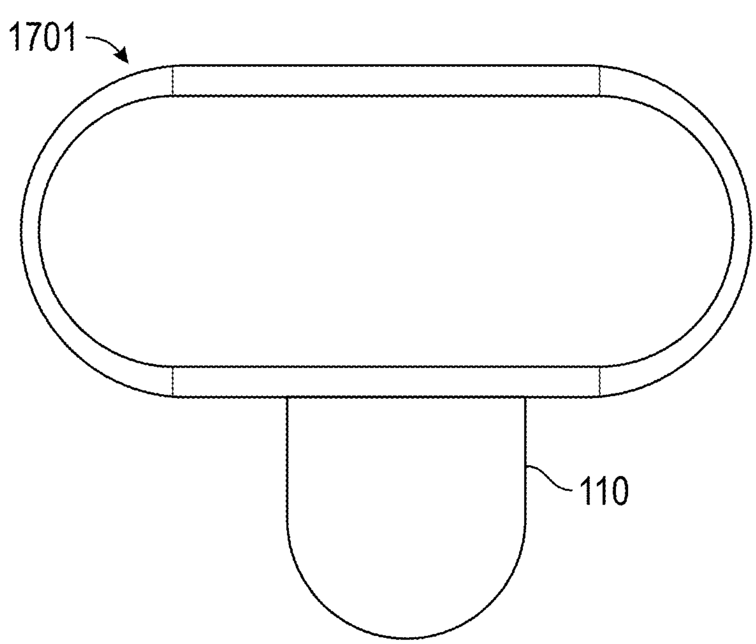

FIG. 23D

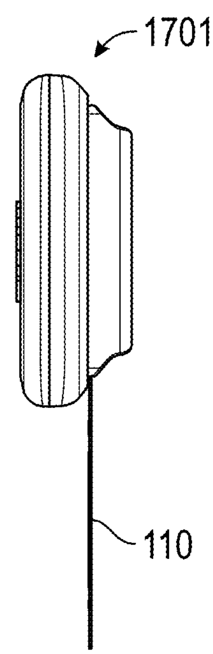

FIG. 23E

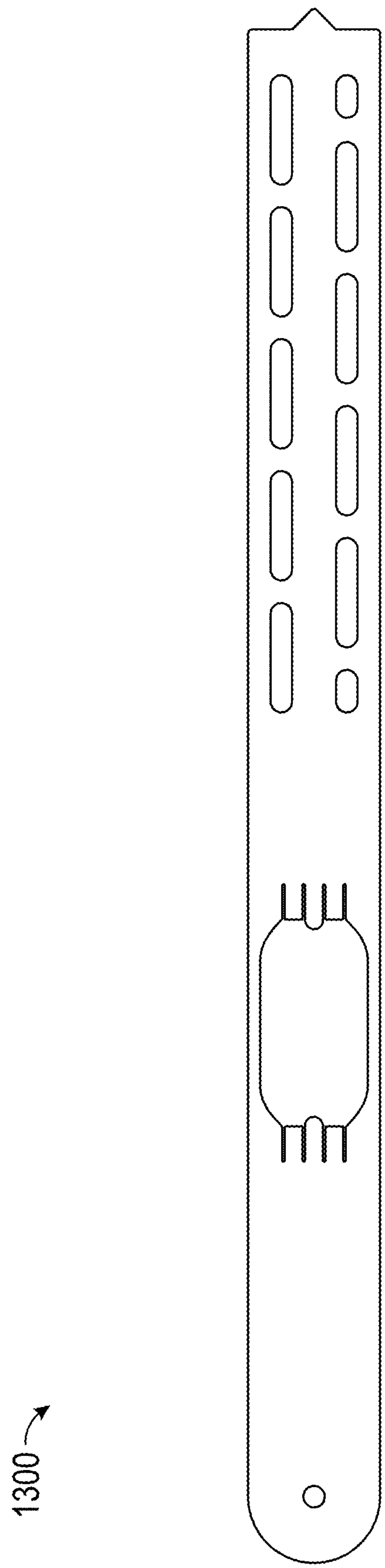

FIG. 24

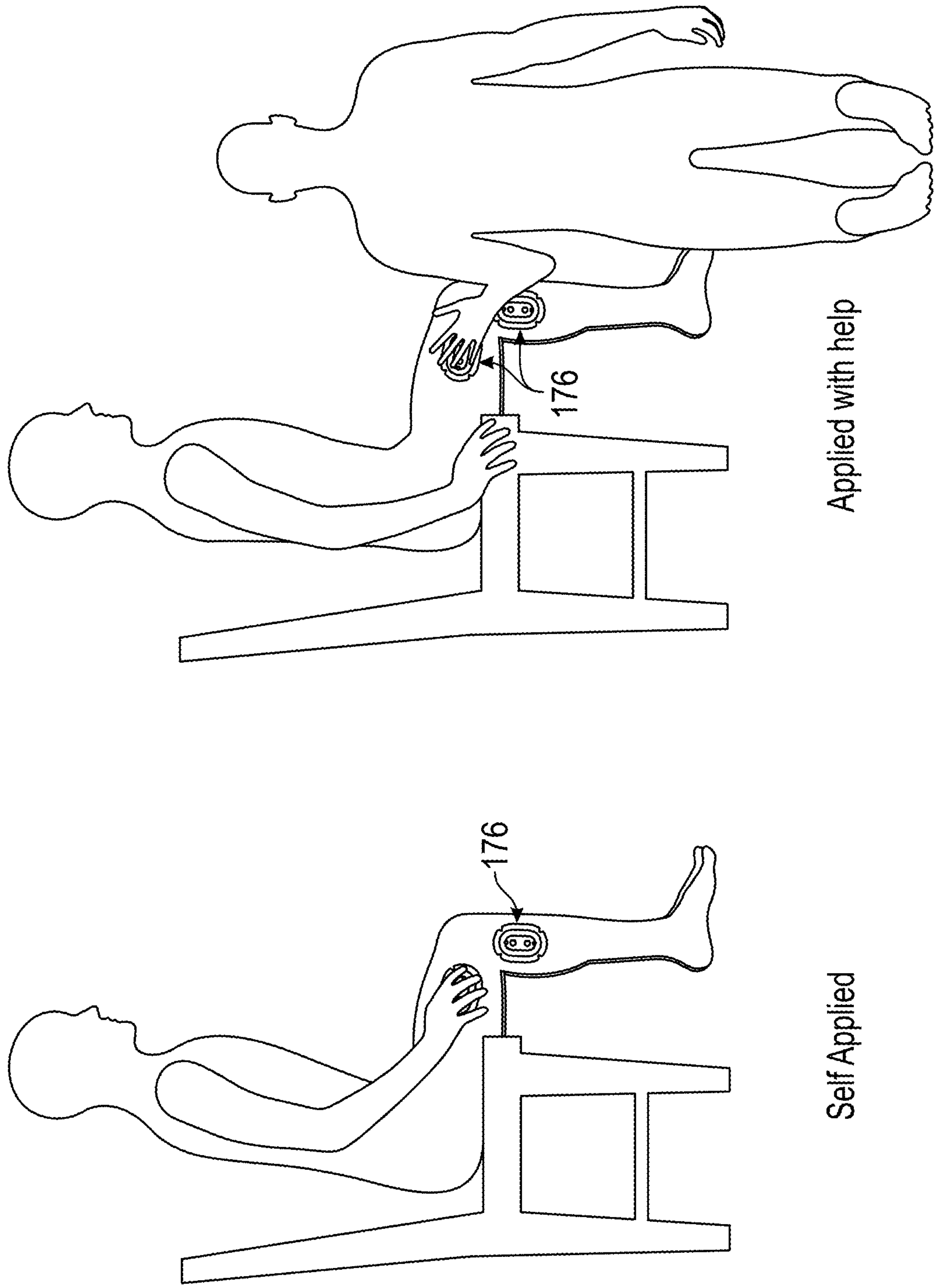

FIG. 25

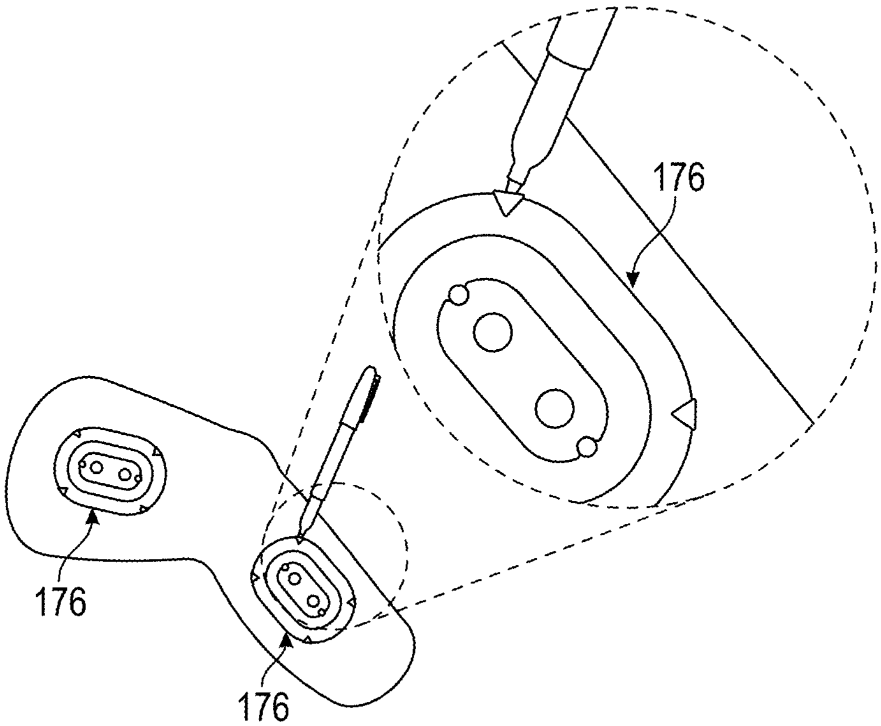

FIG. 26

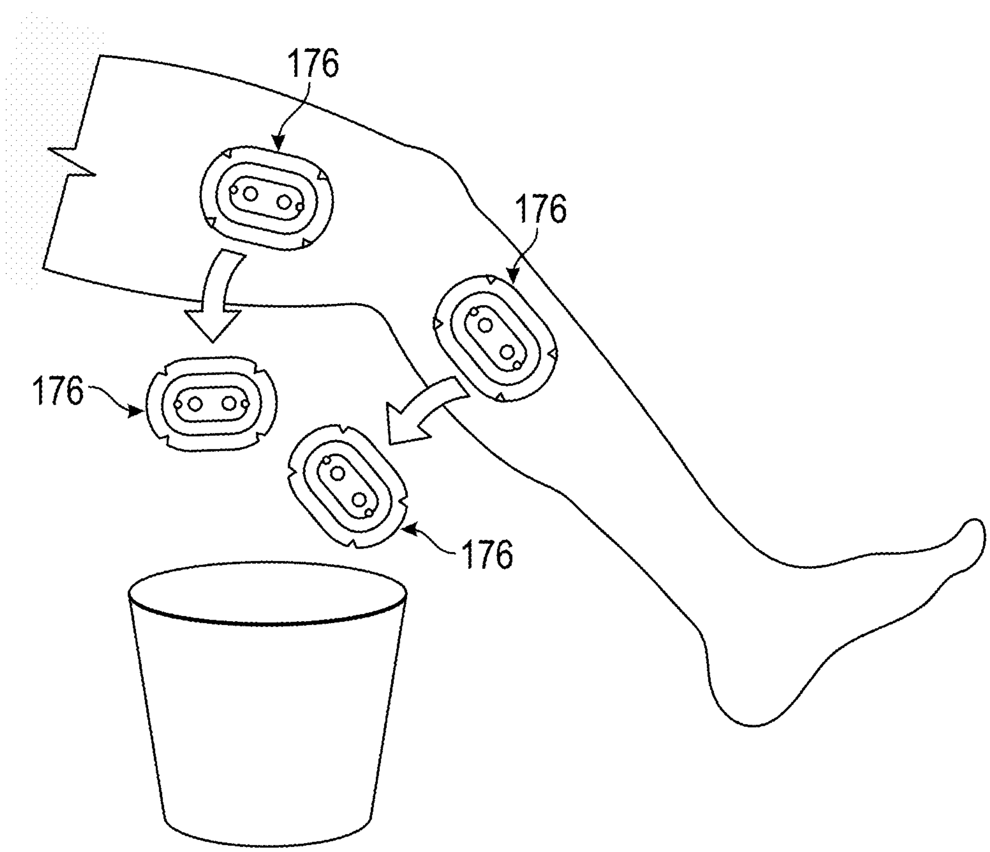

FIG. 27

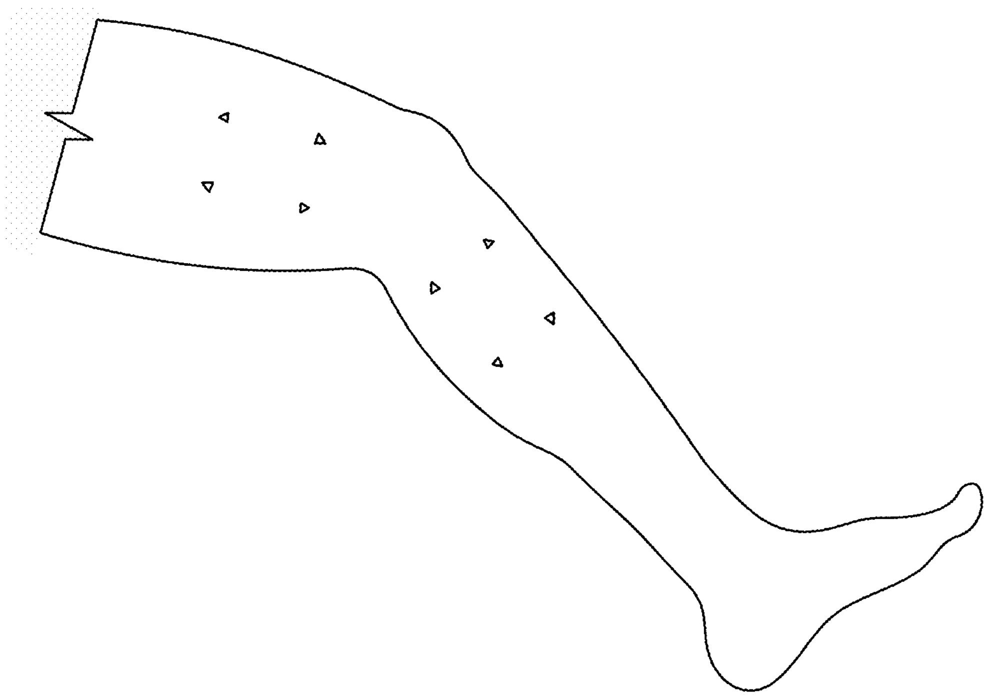

FIG. 28

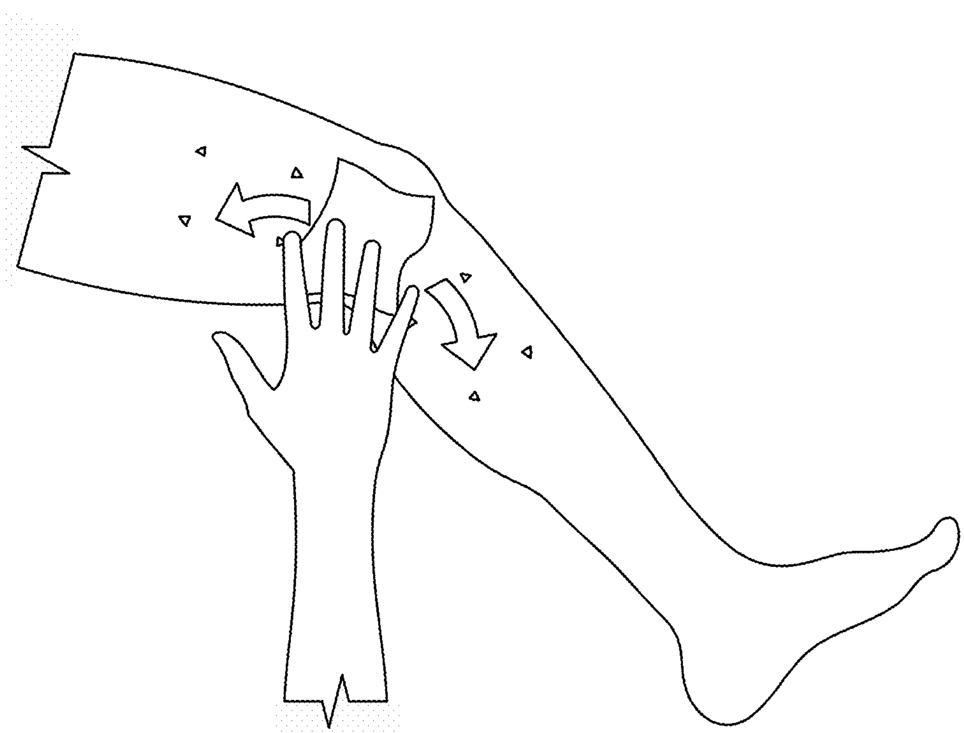

FIG. 29

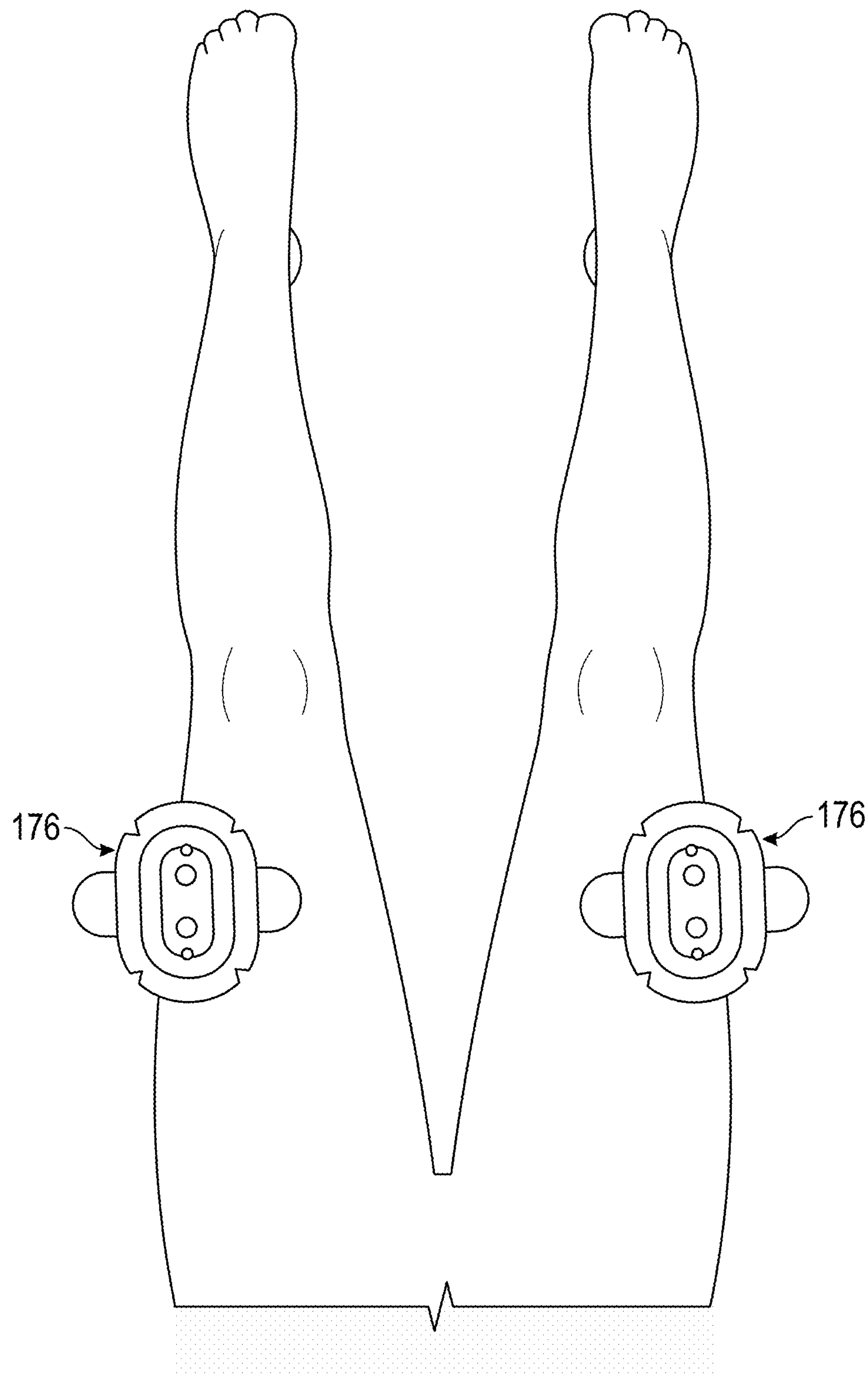

FIG. 30

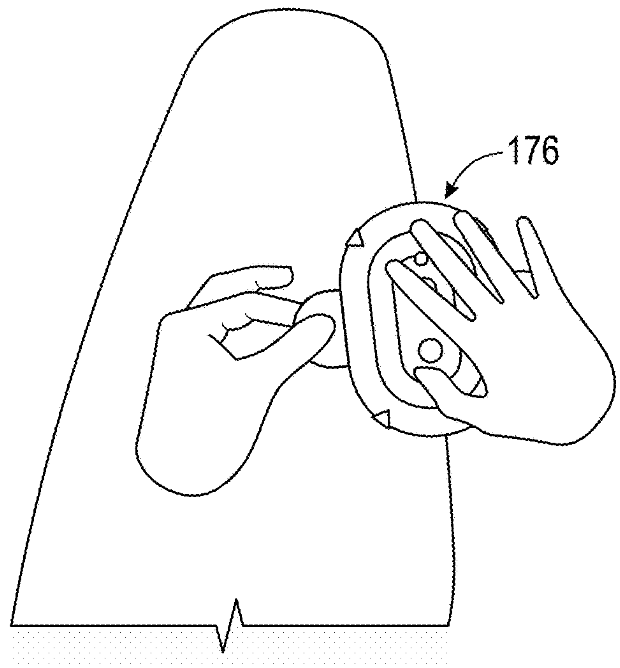

FIG. 31

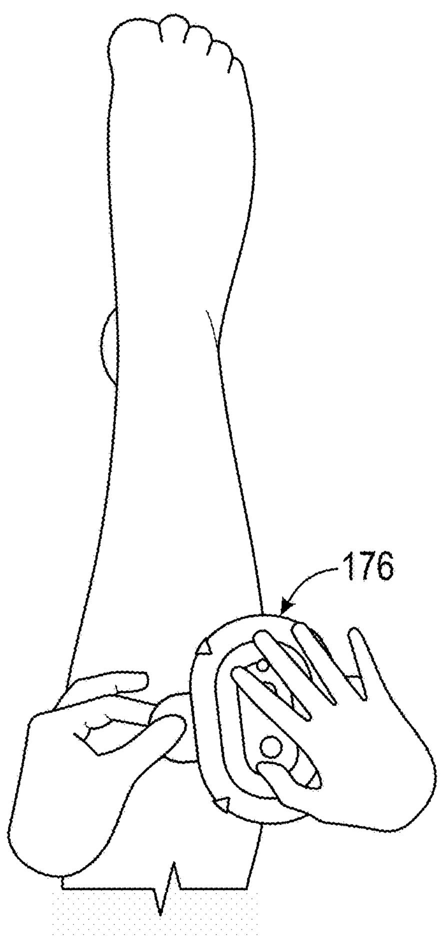

FIG. 32

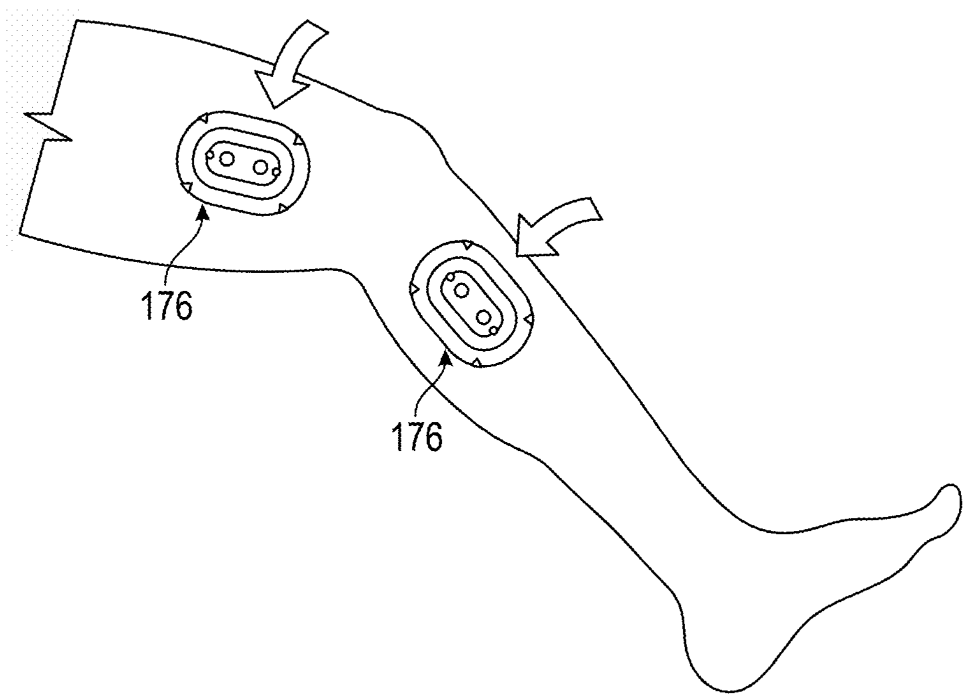

FIG. 33

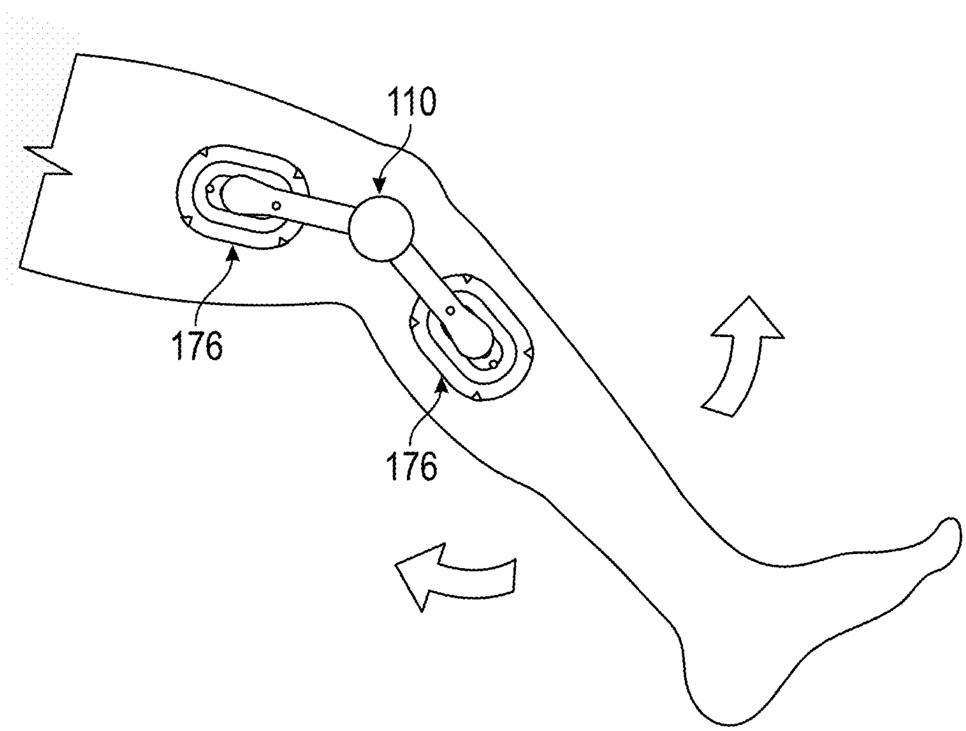

FIG. 34

WEARABLE DEVICE FOR COUPLING TO A USER, AND MEASURING AND MONITORING USER ACTIVITY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional (and claims priority and benefit) of U.S. patent application Ser. No. 17/017,062, filed Sep. 10, 2020, which claims priority to and the benefit of:

U.S. Prov. Pat. App. No. 62/991,295, filed Mar. 18, 2020,
U.S. Prov. Pat. App. No. 62/911,515, filed Oct. 7, 2019,
U.S. Prov. Pat. App. No. 62/904,013, filed Sep. 23, 2019,
U.S. Prov. Pat. App. No. 62/901,464, filed Sep. 17, 2019, and
U.S. Prov. Pat. App. No. 62/901,411, filed Sep. 17, 2019.

Each of these above-referenced applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

This disclosure generally relates to wearable devices and, in particular, to coupling a wearable device to a user and measuring and monitoring activity of the user.

BACKGROUND

A patient often requires physical therapy to recover from surgery, such as a knee replacement surgery. The physical therapy can include exercise to increase the patient's strength, flexibility and stability. If a patient over-extends his or her muscles or joints, the muscles or joints, surrounding tissues or repaired tissues may become further injured. If a patient does not exercise his or her muscles or joints to gain the appropriate range of motion, the joint may become stiff and require additional surgery. Measuring and monitoring the range of motion during physical therapy can help prevent further injury to the patient and result in a faster recovery time.

A goniometer is an instrument that can be used to measure ranges of motion or joint angles of a patient's body. A standard goniometer consists of a stationary arm that cannot move independently, a moving arm attached to a fulcrum in the center of a body, and the body being a protractor of which 0 to 180 or 360 degrees are drawn. The stationary arm is attached to one limb or part of the patient's body (e.g., a thigh) and the moving arm is attached to another limb or part of the patient's body (e.g., a lower leg). The fulcrum can be a rivet or screw-like device at the center of the body that allows the moving arm to move freely on the body of the device in order for a clinician to obtain a measurement of the angle of movement of the patient's joint (e.g., a knee). The measurements can be used to track progress in a rehabilitation program. Each time a patient has a rehabilitation session, the clinician places and hold, or attaches the goniometer device onto the patient, for example, using straps. The patient may have different clinicians setting up the goniometer device and measuring the joint movement. Based on the experience of the clinician, or other person, the goniometer device may be attached onto different locations on the patient, which can affect the accuracy and reproducibility of the measurements. The accuracy of the repeated measurements may also be compromised due to issues with or the sensitivity of the device.

SUMMARY

Exemplary implementations of a system for measuring an angle of a joint of a user are disclosed. The system can include a center hub, a first arm, a second arm, a magnet, and a sensor. The center hub includes a first hub and a second hub. The first arm is configured for attachments to a first limb portion of the user at a first outer end and to the first hub at a first inner end. The second arm is configured for attachments to a second limb portion of the user at a second outer end and to the second hub at a second inner end, wherein the first hub is pivotally coupled to the second hub. The magnet is coupled to the second hub. The sensor is disposed in the center hub and configured to detect a rotation of the magnet.

Other implementations of a system for measuring an angle of a joint of a user can include a center hub, first and second coupling apparatuses, and first and second arms. The center hub has a first hub and a second hub. The first arm is configured for attachment to a first limb portion of the user at a first outer end and to the first hub at a first inner end. The second arm is configured for attachment to a second limb portion of the user at a second outer end and to the second hub at a second inner end, wherein the first hub is pivotally coupled to the second hub and configured to rotate 360 degrees about an axis. The system further comprises a magnet, a printed circuit board (PCB), and a sensor. The magnet is coupled to the second hub. The PCB is removably disposed in the center hub. The sensor is coupled to the PCB and configured to detect a rotation of the magnet.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

For a detailed description of example embodiments, reference will now be made to the accompanying drawings in which:

FIGS. **1**A and **1**B are perspective views of a wearable device for measuring and recording movement in accordance with aspects of the present disclosure.

FIGS. **2**A and **2**B are top and bottom perspective views of a goniometer in accordance with aspects of the present disclosure.

FIGS. **3**A and **3**B are top and side views of a goniometer in accordance with aspects of the present disclosure.

FIG. **4** is an exploded view of a goniometer in accordance with aspects of the present disclosure.

FIGS. **5**A-C are side views of the goniometer with its arms rotating in accordance with aspects of the present disclosure.

FIG. **6** is a perspective view of the goniometer with its arms twisting in accordance with aspects of the present disclosure.

FIGS. **7**A-D are top schematic views of the attachment in accordance with aspects of the present disclosure.

FIGS. **8**A-D are views of an attachment in accordance with aspects of the present disclosure.

FIGS. **9**A-D are views of first and second layers of the attachment in accordance with aspects of the present disclosure.

FIGS. **10**A-D are views of a pod of the attachment in accordance with aspects of the present disclosure.

FIGS. **11**A and **11**B are perspective and cross-sectional views of a coupling apparatus in accordance with aspects of the present disclosure.

FIGS. **12**A and **12**B are perspective views of the attachment and a goniometer attachment in accordance with aspects of the present disclosure.

FIG. **13** is a cross sectional view of the goniometer in accordance with aspects of the present disclosure.

FIGS. **14**A-C are perspective and top views of a center hub of the goniometer closed, open, and with components removed in accordance with aspects of the present disclosure.

FIGS. **15**A and **15**B are top and bottom views of a printed circuit board (PCB) in accordance with aspects of the present disclosure.

FIGS. **16**A and **16**B are graphs illustrating test data in accordance with aspects of the present disclosure.

FIG. **17** is a perspective view of the alignment device, with the alignment device shown aligned relative to the center of the joint of the user.

FIG. **18** is a perspective view of the user illustrating the markings on the skin of the user at the center of the joint, and on the opposing limb portions.

FIG. **19** is a perspective view illustrating the attachments being aligned with the markings on the opposing limb portions.

FIG. **20** is a perspective view illustrating the wearable device being coupled to the attachment, and aligned relative to the center of the joint.

FIG. **21** is a perspective view illustrating an alternative embodiment of the alignment device, and illustrating the light beams at the center of the joint and on the opposing limb portions, which are configured to assist the clinician in aligning the attachments and wearable device relative to the center of the joint.

FIG. **22** is a perspective view illustrating yet another alternative embodiment of the alignment device, and illustrating the diodes at the center of the joint, on the opposing limb portions, and on the wearable device.

FIGS. **23**A-**23**E are various views of an embodiment of a pedometer that may be coupled to the goniometer.

FIG. **24** is a plan view of another embodiment of an alignment device.

FIGS. **25**-**34** are schematic sequential images of an embodiment of alignment and installation method.

NOTATION AND NOMENCLATURE

Various terms are used to refer to particular system components. Different companies may refer to a component by different names—this document does not intend to distinguish between components that differ in name but not function. In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices and connections.

The terminology used herein is for the purpose of describing particular example embodiments only, and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections; however, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms, when used herein, do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C. In another example, the phrase “one or more” when used with a list of items means there may be one item or any suitable number of items exceeding one.

Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” “top,” “bottom,” and the like, may be used herein. These spatially relative terms can be used for ease of description to describe one element’s or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms may also be intended to encompass different orientations of the device in use, or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptions used herein interpreted accordingly.

DETAILED DESCRIPTION

The following discussion is directed to various embodiments of the invention. Although one or more of these embodiments may be preferred, the embodiments disclosed should not be interpreted, or otherwise used, as limiting the scope of the disclosure, including the claims. In addition, one skilled in the art will understand that the following description has broad application, and the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to intimate that the scope of the disclosure, including the claims, is limited to that embodiment.

In accordance with aspects of the present disclosure, FIGS. **1**A and **1**B illustrate an exemplary system or wearable device **100** for measuring and recording flexion and extension at a joint **107** of a user **102**. The wearable device **100** comprises first and second coupling apparatuses, or attachments, **112**, **114**, **118**, **120** (hereinafter referred to as first and second attachments **118**, **120**) and may be configured to be removably coupled to the user **102** at opposing limb portions **104**, **106** of the joint **107**. For example, and as illustrated in FIGS. **1**A and **1**B, the first and second attachments **118**, **120** may be coupled to a leg of a user at opposing limb portions **104**, **106**, e.g., thigh **104**, and calf **106**, of the knee or joint **107** of the user **102**. As will be appreciated by those of skill in the art, the first and second attachments **118**, **120** may be coupled to the user at opposing limb portions of any other joint of a patient or user **102**. It is further contemplated that the wearable device **100** may be utilized for measuring flexion and extension of joints in animals, a joint of a robot, or any other desired joint or joint equivalent.

To position the wearable device **100** relative to the joint **107**, a person, such as a clinician, may identify a joint center **108**, where the joint center **108** may be used to align the wearable device **100** to the joint **107**. The clinician may use an alignment device **300** to identify and mark the joint center **108**. For example, the alignment device **300** may be used to mark the skin of the user **102** at the joint center **108** with a marker, pen, or any other desired tool. Further, the alignment device **300** may be used to identify and mark positions at opposing limb portions **104**, **106** for the first and second attachments **118**, **120** relative to the joint center.

With reference to FIGS. **2A-6**, the wearable device **100** comprises an exemplary device or apparatus **110**, such as a goniometer. Hereinafter, the device or apparatus **110** may be referred to as a goniometer **110**. The goniometer **110** is configured to measure the angular flexion and extension at the joint **107** of the user **102**. The goniometer **110** has a top **122**, a bottom **124**, and opposing sides **126**. The goniometer **110** may comprise a center hub **116** aligned coaxially with an axis A, and first and second arms **128**, **130**, wherein the arms **128**, **130** couple to, and are rotatable about, the axis A and the center hub **116**. More specifically, first and second inner ends **132**, **134** of the respective arms **128**, **130** couple to the center hub **116**. The arms **128**, **130** extend outwardly from the center hub **116** to respective first and second outer ends **168**, **170**. In an alternative embodiment, the arms **128**, **130** may be integrally formed with the center hub **116**. Embodiments of the goniometer can enable various rotational or pivotal motions, such as with an axle, a rack and pinion system, etc.

With reference to FIGS. **3**A and **3**B, the goniometer **110** can have a length L that extends from the first outer end **136** to the second outer end **138**. The length L can vary depending on the relative position of the first and second arms **128**, **130**. For example, a maximum length L of the goniometer **110** may be measured when the arms **128**, **130** are positioned opposing one another. Whereas, when the arms **128**, **130** are positioned parallel, and respectively directly above and below one another, a minimum length L of the goniometer **110** may be measured between the outer ends **136**, **138** to an opposite side of the center hub **116**. A width W of the goniometer may be measured as a diameter of the center hub **116**. Further, a height H of the goniometer **110** may be measured from a bottom of the second arm **130** to a top of the first arm **128**, or to a top of the center hub **116**, whichever is greater.

In an exemplary embodiment, the center hub **116** comprises a first or upper hub **146** and a second or lower hub **148**. The hubs **146**, **148** are coaxially aligned with one another, and with axis A. Moreover, the hubs **146**, **148** may be configured to rotate about the axis A for 360 degrees, and relative to one another. Further, each of the hubs **146**, **148** may have a link arm **143** for coupling between the hubs **146**, **148** and the respective arms **128**, **130**. For example, the first arm **128** may be coupled to the link arm **143** of the first hub **146**, and the second arm **130** may be coupled to the link arm **143** of the second hub **148**.

In operation, embodiments of the arms **128**, **130** may rotate, pivot, flex or extend relative to the center hub **116**. This design can account for the complex motion of a joint, slippage of the joint, and the broad range of shapes and sizes of the patient's joint **107**. In addition, this design can maintain the position of the center hub relative to the joint center **108**. Embodiments of the device can enable freedom of motion in many planes but not in the rotational plane of the joint. This enables the device to fit many different people but still make accurate measurements.

More specifically, and as best illustrated in FIG. **4**, the first and second arms **128**, **130** may each include an inner link **142** disposed adjacent to the respective inner ends **132**, **134**, and an outer link **144** disposed between the inner link **142** and the outer ends **168**, **170**. With reference to FIGS. **5**A-C, the inner link **142** may couple to the link arm **143** in order to couple between respective arms **128**, **130** and hubs **146**, **148**. The inner link **142** may be configured to facilitate the pivot, flex, or extension of the respective arm **128**, **130** relative to the center hub **116**. A pin **164** may be used to couple the inner link **142** and the link arm **143** of each arm **128**, **130** to allow for the pivot, flexion, or extension of the respective arms **128**, **130**. The pin **164** may be disposed perpendicular to the length of the arms **128**, **130**.

The outer link **144** may couple to the inner link **142** and respective outer ends **136**, **138**. With reference to FIG. **6**, the outer link **144** may be configured to couple to the inner link **142** to facilitate rotation of the respective arms **128**, **130** relative to the center hub **116**. A screw **162** may be configured to couple the outer link **144** to the inner link **142** to facilitate rotation of the respective arms **128**, **130** about the screw **162**, all relative to the center hub **116**. The screw **162** may align parallel with the length of the respective arm **128**, **130**. It is to be appreciated that the first and second arms **128**, **130** may rotate +/− eighteen degrees, or any other desired amount, in one or more directions. Further yet, and with reference to FIGS. **5**A-C, the outer link **144** may be configured to couple to the respective outer ends **136**, **138** to facilitate further pivot, flexion, or extension of the respective arms **128**, **130** relative to the center hub **116**. A pin **164** may be used to couple the outer link **144** and the respective outer ends **136**, **138** to allow for further pivot, flexion, or extension, of the respective arms **128**, **130** relative to the center hub **116**. The pin **164** may be disposed perpendicular to the length of the respective arms **128**, **130**.

The first and second outer ends **136**, **138** may comprise first and second goniometer attachments **168**, **170**, which may be integral with, or coupled to the respective outer ends **136**, **138**. It is to be appreciated the goniometer attachments **168**, **170** may couple, or be integral with, the arms **128**, **130** at any desired location, or in any desired configuration. The first and second goniometer attachments **168**, **170** may be configured to removably couple with the attachments **118**, **120**. Further, each goniometer attachment **168**, **170** can comprise one or more bosses **200**, and one or more magnets **158** positioned next to the bosses **200** to facilitate the coupling and alignment of the goniometer attachments **168**, **170** and the attachments **118**, **120**. The bosses **200** and magnets **158** further facilitate the alignment of the goniometer **110** relative to the attachments **118**, **120**. The arms **128**, **130** may also include one or more arm alignment holes **140** configured to align with the attachments **118**, **120**, or an alignment mark on a user **102**. The arms **128**, **130** may further have one or more wings **202** that extend from a side **126** of the goniometer **110**, such as from the first or second goniometer attachments **168**, **170**. The wings **202** can be formed from or coupled to the first or second goniometer attachments **168**, **170**. The wings **202** can be a tab or have any other desired shape. The wings **202** may be configured to assist a user in moving the arms **128**, **130** of the goniometer perpendicularly relative to the attachments to facilitate uncoupling the goniometer **110** from the attachments **118**, **120** without uncoupling the attachments **118**, **120** from the user **102**.

With reference to FIGS. **7-9**, the attachments **118**, **120** may comprise first and second layers **172**, **174** and a pod **176** coupled together with one another. When coupled to one another, each of the layers **172**, **174** and the pod **176** may be concentric with one another. The first attachment **118** may be the same as and interchangeable with the second attachment **120**. The first layer **172**, the second layer **174**, and the pod **176** may be generally oval-shaped or any other desired shape. The first layer **172** may be larger than the second layer **174**, and the second layer **174** may be larger than the pod **176**. Further, the first layer **172** may be thinner than the second layer **174**, and the pod **176** can be thicker than the first and second layers **172**, **174**.

The first layer **172** may have a top **182** and a bottom **184**, and may be formed from a pad, coated paper, plastic, woven fabric, latex, or any other desired material. For example, the top **182** may be formed from a pad and the bottom **184** may comprise an adhesive material **236**, such as a medical-grade adhesive or other suitable material. The adhesive material **236** couples to the skin of the user **102** to couple the attachments **118**, **120** to the user **102**. Further, the top **182** may also have an adhesive layer **194**, which may be smaller in area than the first layer **172**. Further, the adhesive layer **194** can be less than or equal to the area of the second layer **174**. The adhesive layer **194** may be ovular in shape, and define one or more notches or voids in an outer periphery. Further, the first layer **172** can define notches **180** in an outer periphery for assisting in aligning the first attachment **118** relative to a predetermined location, or mark, on the user **102**. The notches **180** can be v-shaped or have any other desired shape. Further yet, the first layer **172** may define a pair of voids or alignment holes **178**, which may assist in the alignment of the first attachment **118** relative to a predetermined location, or mark, on the user **102**. The alignment holes **178** and notches **118** of the first layer **172** may be the same as, or aligned with, the notches or voids of the adhesive layer **194**.

The second layer **174** may have a top **186** and a bottom **188**, and may be formed of a foam material or any other desired material. The second layer **174** may couple to the adhesion layer **194** of the first layer **172**. To prevent uncoupling, the foam material of the second layer **174** may dampen forces between the goniometer **110** and the attachments **118**, **120**. The top **186** of the second layer **174** may also have an adhesive layer **195** on an upper surface **238** of the top **186**, which may be smaller in area than the area of the pod **176**. The adhesive layer **195** may have an ovular shape, or any other desired shape. Further, the adhesive layer **195** and the second layer **174** may have one or more holes, or one or more cutouts that may be aligned with the alignment hole **178** of the first layer **172** to align the adhesive layer **195** and the second layer **174** with the first layer **172**. The adhesive layers **194**, **195** can be formed of an adhesive material or any other desired coupling material, such as a hook- or a loop-type material. The first layer **172** can have a length L**1** and a width W**1**. The second layer **174** can have a length L**2** and a width W**2**. The adhesive layer **195** can have a length L**3** and a width W**3**.

As illustrated in FIGS. **10**A-D, the pod **176** has a top **190** and a bottom **192**. The pod **176** can include one or more notches **196**. For example, notches **196** can be located at opposing ends of the pod **176**. The notches **196** can be used for alignment of the first coupling apparatus **112**, the pod **176**, or any other desired feature of the wearable device **100**.

The pod **176** can include one or more magnets **160**. The magnet **160** may be a neodymium magnet or any other desired magnet. The pod **176** can have a recess for housing the magnet **160**. The magnet **160** can be circular or any other desired shape. Two magnets **160** can be disposed in the pod **176** at opposing ends of the pod **176**. The pod **176** can be sized to be received by and detachably coupled to the upper surface **238** of the second layer **174**.

More specifically, the pod **176** has an underside, such as the bottom **192**, which may couple to the upper surface **238** of the second layer **174**. The bottom **192** can have one or more hooks or loops, to couple to the upper surface **238**. Alternatively, the upper surface **238** and the bottom **192** may comprise an adhesive material to facilitate the detachable coupling between the upper surface **238** and the pod **176**. The top **190** of the pod **176** may have one or more recesses **198**. The recess **198** may be ovular in shape or have any other desired shape. The recesses **198** may also have tapered edges to assist a user **102** in uncoupling from the pod **176** by moving the arms **128**, **130** perpendicularly relative to the pod **176**. Two recesses **198** may be formed in the pod **176** at opposing ends or sides of the pod **176**.

With reference to FIGS. **11-12**, the recesses **198** are sized to receive the bosses **200** to align the goniometer **110** relative to the attachments **118**, **120**. Moreover, the recesses **198** are sized to allow slight movement of the bosses **200** to compensate for slight translational movement of the joints **128**, **130** while the goniometer **110** is worn by the user **102**. This movement may reduce the stress between, and prevent uncoupling of, the attachments **118**, **120** to the skin, clothing, brace, or any other desired location on the user **102**.

When the user **102** moves the first or second limb portions **104**, **106**, the first or second arms **128**, **130** move or rotate with the first and second hubs **146**, **148**. The goniometer **110** can measure the rotation of joint **107** by measuring the angle between the first and second hubs **146**, **148**. To achieve this, and with reference to FIGS. **4**, **13-15**, the center hub **116** defines an opening for receiving and containing a printed circuit board (PCB) **152**, a sensor **216**, a retaining ring **154**, a magnet **156**, or any other components of the goniometer **110** which cooperate with one another to measure relative angular movement between the arms **128**, **130**. More specifically, the first and second hubs **146**, **148** may define the opening of the center hub **116**.

For enclosing the opening, a cover **150** may be attached to the center hub **116**, and more specifically to the first hub **146**. The cover **150** can be detachably coupled to the first hub **146** or any other desired location. The cover **150** can also be configured to inhibit movement of the PCB **152** and other components located within the center hub **116**. For example, when the cover is closed, a bottom portion of the cover **150** may apply direct or indirect pressure to the PCB **152**. The cover **150** may have a snap mechanism **226**, such as a finger snap or any other desired mechanism, configured to attach and detach the cover to the center hub **116**.

The magnet **156** may couple to the second hub **148**, and the sensor **216** also disposed in the center hub **116** is configured to detect rotation of the magnet **156**. The sensor **216** can be configured to measure the rotation of the magnet **156** to a sensitivity up to one-hundredth of a degree, or to any other desired sensitivity.

As illustrated in FIGS. **13-14**C, the center hub **116** includes the first hub **146** positioned rotatably above the second hub **148**. An outward notch **208** may be coupled to the first hub **146**, or any other desired location. The outward notch **208** may be formed with the first hub **146** or attached to the first hub **146**. The PCB **152** may be removably disposed in the center hub **116**. The PCB **152** may have an inward notch **210**. The outward notch **208** may be configured to couple with the inward notch **210** to align the sensor **216** and the magnet **156**. The alignment can restrict movement of the PCB **152** within the center hub **116**.

When the cover **150** is removed, the PCB **152** may be accessed. FIG. **14**B illustrates a battery housing **206** coupled to the PCB **152**. The battery housing **206** may be attached to a top side **212** of the PCB **152**. The battery housing **206** may be formed of conductive metal or any other desired material. When the cover **150** is attached to the first hub **146**, the cover **150** may apply pressure to the battery housing **206**, which may secure PCB **152** within the center hub **116**. A battery **204** may be detachably coupled to the battery housing **206**. The battery **204** may be a lithium ion battery or any other desired battery or power source. The battery housing **206** may have tabs or any other desired contact points for conduction with the battery **204**. The battery **204** may be removed from the battery housing **206** to turn the goniometer **110** off or for replacement of the battery **204**. For example, when the cover **150** is opened or removed from the first hub **146**, the PCB **152** may be removed from the center hub **116** and the battery **204** may be replaced. To maintain calibration of the goniometer **110** after a battery replacement, the inward notch **210** of the PCB **152** is configured for receiving the outward notch **208** of the first hub **146**.

In FIG. **14**C, the cover **150** and the PCB **152** are removed from the center hub **116**. As shown, the second hub **148** may include a magnet housing **224**. The magnet housing **224** may be located in the center of the second hub **148** or any other desired location. The magnet housing **224** may be coupled to the PCB **152** using a weld, an adhesive, a tack, or any other desired attachment. The magnet housing **224** may be configured to receive the magnet **156**. The magnet **156** may be configured to have north and south polarity within the center hub **116** or any other desired polarity. The magnet housing **224** may include a lip at a top of the magnet housing **224** or have any other desired configuration. The retaining ring **154** may be configured to couple the magnet **156** to the second hub **148**. The retaining ring **154** may be coupled to the second hub **148** around the magnet housing **224**. For example, the retaining ring **154** may be positioned between the lip or top of the magnet housing **224** and the second hub **148**. The retaining ring **154** may secure the magnet within the magnet housing **224**. The retaining ring **154** may move with the second hub **148** and the magnet **156** as the second hub **148** rotates with the second arm **130**.

FIG. **15**A illustrates the top side **212** of the PCB **152**. As described above, the battery housing **206** may be attached to the top side **212**. FIG. **15**B illustrates a bottom side **214** of the PCB **152** in accordance with aspects of the present disclosure. The PCB **152** includes components coupled to the bottom side **214**. The components may comprise a circuit **220** including resistors, LEDs, transistors, capacitors, inductors, transducers, diodes, switches, the sensor **216**, a transmitter **222**, or any other desired component. The components may be attached to the PCB **152** using a surface mount method, a through-hole method, or any other desired method. The PCB **152** may include additional and/or fewer components and is not limited to those illustrated in FIGS. **15**A and B.

The circuit **220** may be configured to generate an electrical signal based on the rotation of the magnet **156**. The circuit **220** may be configured to transmit the electrical signal in real time. The circuit **220** may transmit the electrical signal. For example, a transmitter **222** may be coupled to the PCB **152** and configured to transmit an electrical signal based on the rotation of the magnet **156** to an external device. The transmitter **222** may include wired or wireless transmission, such as Bluetooth™, WiFi, NFC or any other means or method of desired transmission. The external device may be a mobile phone, a computer, a tablet, or any other desired device. The external device may have a user interface. The user interface may be configured to receive the electrical signal and display data obtained from the electrical signal. The data may include the angle of the joint **107**, or any other desired information.

The user interface may include an app that receives the data, manipulates the data, records the data, and displays aspects of the data. For example, the app may display the angle of the joint **107** of a user **102**, a history of the angle of the joint **107**, duration of the angle, or any other desired information, such as a measurement of the angle in real time.

The sensor **216** may be a Hall Effect sensor, or any other desired sensor (e.g., a magnetic position sensor AS5601 using internal MEMS Hall Effect sensors). The sensor **216** may be coupled to the PCB **152** or any other desired device. The sensor **216** may be coupled to the bottom side **214** of the PCB **152** at a location directly above the magnet **156** when the PCB **152** is disposed within the center hub **116**. The PCB **152** and the sensor **216** may rotate with the first hub **146** and the first arm **128**. The magnet **156** may rotate with the second hub **148** and the second arm **130**. The design of the wearable device **100**, including the configuration of the sensor **216** and the magnet **156**, may improve the accuracy of the measurements of the angle of the joint **107**.

FIG. **16**A is an exemplary graph **228** with line **232** depicting, while the wearable device **100** is attached to the user **102**, the accuracy of measurements of angles of the joint **107** by the wearable device **100**. FIG. **16**B is an exemplary graph **230** with line **234** depicting the accuracy of measurements of angles of the joint **107** by a different measurement device attached to a user using Velcro™ straps. The standard deviation of the measurements made by the different measurement device shown by line **234** is greater than the standard deviation of the measurements made by the wearable device **100** shown by line **232**. Users, such as clinicians or patients, are able to more accurately initially place and realign the first and second attachments **118** and **120** of the wearable device **100** as compared to using the Velcro™ straps. The configuration of the first and second arms **128**, **130** of the wearable device **100** to fit different users may increase accuracy of the measurements. The configuration of the components, including the PCB **152**, the sensor **216**, and the magnet **156** within the center hub **116** of the wearable device **100** may further increase accuracy of the measurements. The wearable device **100** may be configured to measure the angle of the joint **107** up to an accuracy of measurement up to a one hundredth degree. The different measurement device may have an accuracy up to five degrees. In other words, the measurements taken by the different measurement device may have an accuracy of about +/− five degrees, such as about +/1 1 degree, or even +/− about 0.01 degree from the actual angle of the joint **107**.

With reference to FIGS. **17** and **18**, the alignment device **300** may be used to assist a clinician to align the wearable device **100** (see FIG. **20**) on opposing limb **104**, **106** portions of the joint **107** of the user **102** relative to a center of the joint **107**. Properly aligning the wearable device **100** relative to the joint **107** facilitates greater accuracy and precision of the measurements made with the wearable device **100**. Hence, the need for a device, such as alignment device **300**, to assist in aligning the wearable device **100** relative to the joint **107** of a patient **102**.

The alignment device **300** comprises a first segment **302** and a second segment **304** pivotably coupled to the first segment **302** at a pivot point P. More specifically, the segments **302**, **304** each have a coupling end **306** where the segments **302**, **304** pivotably couple to one another at the pivot point P. Moreover, the pivot point P is spaced adjacently and equidistant from each coupling end **306** of the segments **302**, **304**. Further yet, when the segments **302**, **304** are pivotally coupled, center axes C of each segment **302**, **304** intersect at the pivot point P.

The alignment device **300** further includes voids **308** defined by each segment **302**, **304**, and the voids **308** facilitate the marking of the skin of the user **102** during use of the alignment device **300**. The voids **308** are spaced equidistant on each segment **302**, **304** from the pivot point P, and the spacing between, and size, of each void **308** is commensurate with the spacing between, and size, of each alignment hole **178** of the attachments **118**, **120** (see FIGS. **19** and **20**). Further yet, when the attachments **118**, **120** are coupled to the goniometer attachments **168**, **170**, the spacing between, and position of, each void **308** and the pivot point P is commensurate with the spacing between the center hub **116** and the alignment holes **178** of the attachments **118**, **120**.

Using the alignment device **300**, the clinician may locate the center of the joint **107** of the user **102**, and then position the pivot point P at the center of the joint **107**. When the joint **107** of the user **102** is a knee, the clinician may locate the lateral epicondyle of the knee, and position the pivot point P adjacent to the lateral epicondyle. With the pivot point P positioned adjacent to the lateral epicondyle, the clinician may centrally position each segment **302**, **304** adjacent to the opposing limb portions **104**, **106** of the joint **107**. The clinical may rely on the center axes C of each segment **302**, **304** to assist in centrally aligning with the segments **302**, **304** to the opposing limb portions **104**, **106**. With reference to FIGS. **17** and **18**, with the segments **302**, **304** centrally aligned with the opposing limb portions **104**, **106**, the clinician may use the voids **308** to mark the skin of the user **102**. Thereafter, and with reference to FIGS. **19** and **20**, the clinician may coaxially align the alignment holes **178** of the attachments **118**, **120** with the markings on the skin of the user **102** to facilitate the alignment, and coupling to the user **102**, of the wearable device **100** relative to the center of the joint **107**. Further, the clinician may use pegs to assist in aligning with the attachments **118**, **120** coaxially with the marks.

Alternatively, and with reference to FIGS. **21** and **22**, an alignment device **400** may be used to assist a clinician to facilitate the alignment of the wearable device **100** relative to a center of the joint **107** of the user. More specifically, alignment device **400** may rely on data from an imaging device **402** of the joint **107** to assist the clinician in positioning the wearable device **100** on the user **102**. The imaging device **402** may be an x-ray, ultrasound, any other imaging device capable of providing image data in 2D, 3D or 4D to the alignment device **400**, or any other imaging device. A processor **404** of the alignment device **400** receives and processes the image data from the imaging device **402**, and the processor **400** facilitates communication from the alignment device **400** to the clinician for the proper positioning of the attachments **118**, **120**, or the wearable device **100**.

In one embodiment, and with reference to FIG. **21**, the alignment device **400** may include one or more lasers **406** that produce a light beam. In this embodiment, the processor **404** may communicate with the lasers **406** to cause the lasers **406** to be positioned such that light beams causes spots of light on the skin of the user **102** on the center of the joint **107** and on opposing limb portion **104**, **106**. The spacing between, and size of, the spots of light are commensurate with the spacing between, and size of, each alignment hole **178** of the attachments **118**, **120**. The clinician may rely on the spots to align the wearable device **100** to center of the joint **107**. Alternatively, the clinician may mark the locations of the spots of light on the skin, similar to the markings used with alignment device **300**, and rely on the markings to align the wearable device **100** to the center of the joint **107**. Further yet, the wearable device **100** may have indicia which the clinician may align with the light beams to facilitate alignment of the wearable device **100**. In yet another alternative, the light beam may create an outline of the wearable device **100** on the skin of the user **102**, where the outline may be commensurate to a perimeter of the wearable device, and may be used to properly align the wearable device **100** to the center of the joint **107**.

In another embodiment of the alignment device **400**, and with reference to FIG. **22**, to assist the clinician in aligning the wearable device **100** relative to a center of the joint **107** of the user **102**, the imaging device **402** may communicate with diodes **408** positioned on the user **102** and/or the wearable device **100**. The diodes **408** may provide position data to the processor **404** such that the processor **404** may communicate with, and position, the lasers **406** to assist the clinician in the positioning of the attachments **118**, **120**. Alternatively, with the position data of the diodes **408**, the processor **404** may communicate, to the clinician, with a clinician communication device **410**, to assist in the alignment of the wearable device **100**. Further yet, the wearable device **100** may also have diodes **408**, which may communicate with the imaging device **402**. Relying on position data of the diodes **408** from the imaging device **402**, the processor **404** may cause the speaker **412** or the display **414** to provide respective audio or image outputs to communicate instructions, to the clinician, to facilitate the alignment of the wearable device **100**. For example, the speaker **412** may provide audio guidance to the clinician, such as "move wearable device 5 mm to the left," to assist in the positioning of the wearable device **100**. In another example, the display **414** may provide visual guidance to the user **107** on the location of the wearable device **100**, and instructions for adjusting its position to achieve proper alignment. The display **414** may be a monitor, or iPad, or any other device for providing an image to the user, such as 3D googles. Of course, any of the above techniques could be used in combination with one another to assist the clinician to properly align the wearable device **100**.

FIGS. **23A-23E** depict an embodiment of an ambulation monitor or pedometer **1701**. Versions of the pedometer **1701** can count the number of steps that a user takes, such as a daily count of steps post-operatively. Such a device can be carried by the user or attached to the user or to a peripheral of the user, such as the goniometer **110**. The pedometer **1701** is operational to track steps of the user even if the user requires a walker or other assistance device.

In some versions, the pedometer **1701** can include the ability to attach to the magnets of the goniometer **110** to ensure accurate tracking of all steps of the user. The pedometer **1701** can include metallic elements that are magnetically attracted to the magnets of the goniometer **110**. Alternatively, additional magnets **1703** mounted to the pedometer **1701**. Embodiments of the pedometer **1701** can further include a body **1705**, a removable cap **1707**, a circuit board **1709** including one or more sensors (e.g., a motion sensor such as an accelerometer, mechanical sensor or other electromechanical sensor), a battery **1711** and fasteners **1713**.

FIG. **24** is a plan view of another embodiment of an alignment device **1300**. Alignment device **1300** can be used in manners that are similar to the previously described alignment devices.

FIGS. **25**-**34** are schematic sequential images of an embodiment of alignment and installation method. As shown in FIG. **25**, the patient or user can change their own pods **176**. Alternatively, another person can help the user change their pods **176**. Although these images are shown for the user's right leg, the actions are mirrored for the user's left leg.

Before either pod **176** is removed (FIG. **26**), a permanent ink marker can be used to mark the user's skin with four marks (e.g., triangles) inside the four V-shaped notches on each of the user's current pods **176**. A total of eight triangular marks can be made, including four marks on the pod attached to the user's upper leg and four marks on the pod attached to the user's lower leg. These marks can be used to correctly position the new pods **176**.

Starting at the edge of one of the pods **176** (FIG. **27**), the adhesive portion of the pod can be gently peeled away and off of the user's skin. Repeat for the second pod **176**, and discard both of the old pods **176**. After removing both old pods **176** from the user's skin (FIG. **28**), the user can wait 24 hours before putting on new pods **176**. This gives the skin time to "air out" and restore itself after being covered for a week. The area may be cleaned with mild soap to avoid removal of the ink markings. If one or more of the markings are fading, they can be re-marked before they disappear entirely.

In FIG. **29**, the area is cleansed with a simple hand soap, rinsed and dried. In FIG. **30**, two new pods **176** are removed from their packaging and correctly oriented such that the "1" tab is pointing toward the center of the user, and the "2" tab is pointing outward away from the user. In FIG. **31**, the V-notches in the pod are aligned with the triangles drawn on the user's upper leg. The same hand as the leg working on is used to hold the center of the pod **176** onto the leg so that the V-notches are aligned with the triangle marks. With the other hand, the tab "1" is gently pulled to expose the adhesive on the back of the pod **176**. The center of the pod **176** is kept in place as the backing paper is removed. Once the "1" backing paper is removed, the adhesive flap is rolled down so that it sticks to the skin smoothly and without getting wrinkled. The V-notches should still be aligned with the triangle marks. The adhesive is gently pressed onto the skin, making sure it has fully adhered to the skin. Next, the "2" tab is removed from the rest of the backing paper. The rest of the adhesive is rolled onto the skin. The entire adhesive can be smoothed to make sure it has fully adhered to the skin.

In FIG. **32**, another pod **176** is oriented on the lower leg so that the "1" tab is pointing toward the center of the body and the "2" tab is pointing away from the body. Next, the V-notches in the pod are aligned with the triangle marks drawn on the lower leg. Using the same hand as the leg being worked on, the center of the pod **176** is held onto the leg so that the V-notches are aligned with the triangle marks. With the other hand, the "1" tab is removed to expose the adhesive on the back of the pod **176**. The center of the pod **176** should be held in place as the backing paper is removed. Once the "1" tab backing paper is removed, the adhesive flap is rolled down so that it sticks to the skin smoothly and without getting wrinkled. The V-notches should still be aligned with the triangle marks. Gently press the adhesive onto the skin, making sure it has fully adhered to the skin. The "2" tab is gently removed to roll the rest of the adhesive onto the skin. Smooth down the entire adhesive to make sure it has fully adhered to the skin.

The fingertips can be used to flatten the edges of the adhesive fully against the skin (FIG. **33**). Rub gently to increase adhesion for both pods **176**. The pods **176** are now changed and the goniometer **110** is ready to attach to the user. Hold the goniometer **110** close to the magnets on the pods **176** and let them snap into place via the magnets. Positioning of the pods **176** can be checked by attaching the goniometer **110** and flexing the leg (FIG. **34**). If the goniometer **110** pops off of one or both of the pods **176**, the detached pod **176** has been re-applied too far from the correct position. Check to see if any V-notches are not well aligned with the triangle marks on the skin. If this is the case, remove that pod **176** and reapply a new pod **176**, making sure it is aligned more closely to the triangle marks on the skin.

Pod Function

A pair of truncated pull tab pod assemblies, one adhered to the upper leg, and one to the lower leg, act as anchors to the ends of the goniometer, which measures the angle of the knee over the course of the patient's sessions. Although the goniometer is removed between sessions, the pair of pods are to be left in place for about a week. Once a week, the patient or caregiver replaces the first set of pod assemblies with new pod assemblies. Repeat weekly. Location targets marked on the legs can be "refreshed" by the patient to provide for repeatable placement locations for each new pair of pod assemblies. The "semi-continuous" measurement anchor points provided by the pods allow for a reliable assessment of range of motion over the duration of the therapy.

Pod Description

The pod assembly can include three basic layers. The bottom layer can be highly compliant, hypoallergenic, breathable, and adheres to the skin. The middle layer can be a thin foam that is moderately compliant. The top layer can be a rigid plastic molded part that has locating features and a pair of magnets and is adhered to the middle layer with an adhesive.

Four location notches in the bottom layer can provide for reliable marking of initial pod locations such that subsequent pods can be placed accurately in the same location and orientation. The middle layer can act as a transitional material between the larger, compliant layer and the smaller pod with snaps. Features in the rigid pod can include the top layer as location features to assemble to the goniometer and to the pod template. The magnets, mating with similar magnets in the goniometer, provide the retention force to allow use of the goniometer during a session. Before assembly to the body, the pod comes from the package with two release papers with large pull tabs that cover the adhesive of the bottom layer.

Pod Design Aspects

The pods can be consistent anchor points for the goniometer. When the goniometer is not in place during a session, either the pods can act as attachment points for the goniometer or a pedometer. For the goniometer to provide relevant and reliable range of motion information for clinical use, the pods can be initially located accurately with respect to physiological landmarks and reliably replaced. The pod assembly allows for accurate initial placement and reliable replacement. A combination of novel features and functions in the pod assembly and the alignment assembly allow for these features. Within the pod assembly, the V-notches in allow for accurate replacement of subsequent pods. The asymmetric release paper adhesive backing and large numbered pull tabs allow for the easy location and highly reliable placement and adhesion to the skin. Once the skin is marked with the four targets triangles, the patient can easily locate and orient a new pod by aligning the V-notches to the marked triangles. This "peel-in-place" feature allows for targeting and adhering to be decoupled. Once the first section of the adhesive is revealed and adhered to the skin it acts as an anchor for the following steps. The patient or caregiver can pull the second pull tab and reveal the rest of the adhesive, and can simply smooth the rest of the pod onto the skin with little concern it will be out of place.

Alignment Assembly Function

The alignment assembly allows physiological landmarks of the patient to be used for accurate pods placements for use with the goniometer. The skin over the lateral epicondyle is located and a circular sticker is placed on the patient's skin or clothing over both the greater trochanter and lateral malleolus. A knee pivot anchor of the alignment assembly can be accurately placed over the lateral epicondyle. The pod is placed with the alignment assembly such that it "points" to either the greater trochanter or lateral malleolus target sticker. Once aligned, the assembly is stabilized and affixed. The process is repeated for other pods as described herein.

Alignment Assembly Description

The alignment arm can be a long chipboard part that has a hole to accept the knee pivot anchor, "pointing" features at the far end to help align with the physiological landmarks, and a hole in the mid-section that aligns to and snaps to features on pods. The distance between these two holes in the template assists in final placement of the pods, and corresponds to hard dimensions in the goniometer. The knee pivot anchor can be a plastic part with a pad of adhesive that snaps into a hole in the end of the template and allows for rotation such that it can be used for both segments of the leg. The center of the anchor can be hollow so that a mark can be placed over the lateral epicondyle. The "skin-side" of the anchor is designed to be flexible, such that the adhesive pad assembled to it can conform to the side of the knee and provide a good adhesion point for use. This part can be removed.

These designs provide accuracy, repeatability, and ease of use for both the patient and the caregiver. The "peel-in-place" release paper design can be superior to the conventional "peel then place" method for the pods. The "mark after placement" method for initial alignment rather than conventional "mark (with a template) then place" method also is an advantage. The release paper can be asymmetric to allow for holding during the first release paper pull. The release paper can use pull tabs to number the release paper in the order of release. In addition to being an ergonomic aid (a "third hand") for stabilizing the assembly to the leg, the locked pivot point, used for both pod placements insures that the pod-to-pod distance will be well controlled, and within the tolerances of use with the corresponding and highly repeatable goniometer geometry.

Other embodiments can include one or more of the following items or components.

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A system for measuring an angle of a joint of a user, comprising:

- a center hub, the center hub having a first hub and a second hub;
- a first arm configured for attachment to a first limb portion of the user at a first outer end and to the first hub at a first inner end;
- a second arm configured for attachment to a second limb portion of the user at a second outer end and to the second hub at a second inner end, wherein the first hub is pivotally coupled to the second hub;
- a magnet coupled to the second hub; and
- a sensor disposed in the center hub and configured to detect a rotation of the magnet.

The system further comprising a printed circuit board (PCB) removably disposed in the center hub, the PCB having an inward notch; and an outward notch coupled to the first hub, wherein the outward notch is configured to couple with the inward notch to align the sensor and the magnet.

The system further comprising a cover detachably coupled to the first hub, wherein the cover is configured to inhibit movement of the PCB within the center hub.

The system wherein the sensor is a Hall Effect sensor coupled to the PCB.

The system wherein the sensor is configured to measure the rotation of the magnet up to a precision of about +/−0.01 degree.

The system further comprising a retaining ring configured to couple the magnet to the second hub.

The system wherein the magnet is configured with north and south polarity within the center hub.

The system further comprising a circuit configured to generate an electrical signal based on the rotation of the magnet and to transmit the electrical signal in real time.

The system further comprising a user interface configured to receive the electrical signal and display data obtained from the electrical signal.

The system wherein the data includes the angle of the joint of the user.

The system wherein the first hub and the second hub are configured to rotate 360 degrees about an axis.

A system for measuring an angle of a joint of a user, comprising:

- a center hub, the center hub having a first hub and a second hub;
- a first arm configured for attachment to a first limb portion of the user at a first outer end and to the first hub at a first inner end;
- a second arm configured for attachment to a second limb portion of the user at a second outer end and to the second hub at a second inner end, wherein the first hub is pivotally coupled to the second hub and configured to rotate 360 degrees about an axis;
- a magnet coupled to the second hub;
- a printed circuit board (PCB) removably disposed in the center hub; and
- a sensor coupled to the PCB and configured to detect a rotation of the magnet.

The system further comprising a battery housing coupled to the PCB; and a battery detachably coupled to the battery housing.

The system further comprising a transmitter coupled to the PCB and configured to transmit an electrical signal based on the rotation of the magnet.

The system further comprising a user interface configured to receive the electrical signal and display data obtained from the electrical signal, wherein the data has the angle of the joint of the user.

The system wherein the first and second arms are adhesively attached to respective ones of the first and second limb portions of the user.

A system for measuring an angle of a joint of a user, comprising:

a center hub, the center hub having a first hub and a second hub, wherein the first hub has an outward notch;

a first and second coupling apparatus configured to be adhesively attached to first and second limb portions proximal to the joint of the user;

a first arm configured for attachment to a first coupling apparatus at a first outer end and to the first hub at a first inner end;

a second arm configured for attachment to a second coupling apparatus at a second outer end and to the second hub at a second inner end, wherein the first hub is pivotally coupled to the second hub;

a magnet coupled to the second hub;

a printed circuit board (PCB) having an inward notch and removably disposed in the center hub;

a sensor coupled to the PCB and configured to detect a rotation of the magnet, wherein the outward notch is configured to fit within the inward notch to align the sensor and the magnet; and

a transmitter coupled to the PCB and configured to transmit an electrical signal based on the rotation of the magnet, wherein the electrical signal has data of the angle of the joint of the user.

The system further comprising a cover coupled to the center hub, wherein the cover has a snap mechanism configured to attach and detach the cover.

The system wherein the first and second arms have sections configured to rotate the first and second arms about +/− ten degrees.

The system wherein the first and second arms have sections configured to twist the first and second arms about +/− eighteen degrees.

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An apparatus for coupling a device to a user, the apparatus comprising:

a first layer having a top and a bottom, wherein the bottom comprises an adhesive material configured to couple the device to the user;

a second layer concentric with the first layer and coupled to the top of the first layer, and the second layer is smaller in size than the first layer and has an upper surface area;

a pod concentric with the second layer and sized to be received by and detachably coupled to the upper surface area, and the pod is configured to detachably couple the pod to the device.

The apparatus wherein, to prevent the apparatus from uncoupling from the user, the second layer comprises a material configured to dampen forces acting between the device and the apparatus and caused by relative movement between the device and the apparatus.

The apparatus wherein the adhesive material is a medical grade adhesive.

The apparatus wherein the upper surface area comprises one of hooks and loops, wherein the pod has an underside comprised of one of hooks and loops, and wherein the upper surface and the underside are detachably coupled by the hooks and loops.

The apparatus wherein the first layer defines a notch for assisting in aligning the apparatus relative to a predetermined location on the user.

The apparatus wherein the notch is V-shaped.

The apparatus wherein the first layer defines a pair of alignment holes.

The apparatus wherein, to facilitate alignment of the apparatus relative to a predetermined location on the user, the upper surface of the second layer comprises an adhesive material to enable detachable coupling between the upper surface and the pod.

The apparatus wherein, when the device couples to the apparatus, the pod defines a recess for positioning and aligning the device relative to the apparatus.

The apparatus wherein the pod comprises a magnet positioned adjacent to the recess to detachably couple the pod to the apparatus.

An apparatus for coupling a device to a user, the apparatus comprising:

a first layer with a first periphery, and the first layer has a top and a bottom, wherein the bottom comprises an adhesive material configured to couple the apparatus to the user, and the first layer defines a plurality of notches spaced apart from one another and disposed about the first periphery of the first layer, and the first layer defines a pair of alignment holes, and the notches and the alignment holes align the apparatus relative to a predetermined location on the user;

a second layer smaller in size than, and concentric with, the first layer, and the second layer has a lower surface that couples to the first layer and an upper surface, wherein the upper surface presents an area that comprises an adhesive material;

a pod concentric with the second layer and sized to be received by and detachably coupled to the area of the upper surface area, and the pod has a magnet to allow the pod to be detachably coupled to the device, and the pod defines a recess for positioning and aligning the device relative to the apparatus when the device couples to the pod.

The apparatus wherein, to prevent the apparatus from uncoupling from the user, the second layer comprises a material configured to dampen forces acting between and caused by relative movement between the device and the apparatus.

An apparatus for measuring flexion and extension of a joint of a user, said apparatus comprising:

a first attachment and a second attachment, each comprising:

a first layer with a top and a bottom, wherein the bottom comprises an adhesive material for coupling, on opposite sides of the joint, the attachments to the user;

a second layer concentric with the first layer, and the second layer has a lower surface that couples to the first layer and an upper surface; and

a pod concentric with the second layer and detachably coupled to the upper surface; and

a goniometer for measuring flexion and extension of the joint of the user, and the goniometer is detachably coupled to, and spans between, the pods of the attachments.

The apparatus wherein, to prevent the attachments from uncoupling from the user, the second layer comprises a material configured to dampen forces acting between the goniometer and the attachments and caused by relative movement between the goniometer and the attachments.

The apparatus wherein the goniometer has a pair of arms rotatably coupled to and about a center hub, and each arm has a boss for aligning and coupling each arm to a respective pod.

The apparatus wherein each pod defines a recess for receiving and detachably coupling to one of the bosses of the goniometer, and for positioning the goniometer relative to each attachment.

The apparatus wherein, to prevent the goniometer from uncoupling from the attachments, the bosses are sized to be movable within the recesses to allow movement of the bosses in the recesses.

The apparatus wherein, to detachably couple the goniometer to the attachments, magnets are disposed adjacent to each boss and each recess.

The apparatus wherein wings extend outwardly from the sides of each arm of the goniometer configured to enable a user to move the arms of the goniometer perpendicularly relative to the attachments, and to uncouple the goniometer from the attachments.

The apparatus wherein the edges of the pod are tapered, such configuration enabling a user to further move the arms of the goniometer perpendicularly relative to the attachments, and to uncouple the goniometer from the attachments.

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An apparatus for coupling a device to a user, the apparatus comprising:

- a first layer having a top and a bottom, wherein the bottom comprises an adhesive material configured to couple the device to the user;
- a second layer concentric with the first layer and coupled to the top of the first layer, and the second layer is smaller in size than the first layer and has an upper surface area;
- a pod concentric with the second layer and sized to be received by and detachably coupled to the upper surface area, and the pod is configured to detachably couple the pod to the device; and
- a pedometer configured to count steps of the user and configured to be removably attached to the apparatus.

The apparatus wherein, to prevent the apparatus from uncoupling from the user, the second layer comprises a material configured to dampen forces acting between the device and the apparatus and caused by relative movement between the device and the apparatus.

The apparatus wherein the adhesive material is a medical grade adhesive.

The apparatus wherein the upper surface area comprises one of hooks and loops, wherein the pod has an underside comprised of one of hooks and loops, and wherein the upper surface and the underside are detachably coupled by the hooks and loops.

The apparatus wherein the first layer defines a notch for assisting in aligning the apparatus relative to a predetermined location on the user.

The apparatus wherein the notch is V-shaped.

The apparatus wherein the first layer defines a pair of alignment holes.

The apparatus wherein, to facilitate alignment of the apparatus relative to a predetermined location on the user, the upper surface of the second layer comprises an adhesive material to enable detachable coupling between the upper surface and the pod.

The apparatus wherein, when the device couples to the apparatus, the pod defines a recess for positioning and aligning the device relative to the apparatus.

The apparatus wherein the pod comprises a magnet positioned adjacent to the recess to detachably couple the pod to the apparatus.

An apparatus for coupling a device to a user, the apparatus comprising:

- a first layer with a first periphery, and the first layer has a top and a bottom, wherein the bottom comprises an adhesive material configured to couple the apparatus to the user, and the first layer defines a plurality of notches spaced apart from one another and disposed about the first periphery of the first layer, and the first layer defines a pair of alignment holes, and the notches and the alignment holes align the apparatus relative to a predetermined location on the user;
- a second layer smaller in size than, and concentric with, the first layer, and the second layer has a lower surface that couples to the first layer and an upper surface, wherein the upper surface presents an area that comprises an adhesive material;
- a pod concentric with the second layer and sized to be received by and detachably coupled to the area of the upper surface area, and the pod has a magnet to allow the pod to be detachably coupled to the device, and the pod defines a recess for positioning and aligning the device relative to the apparatus when the device couples to the pod; and
- a pedometer configured to count steps of the user and configured to be removably attached to the apparatus.

The apparatus wherein, to prevent the apparatus from uncoupling from the user, the second layer comprises a material configured to dampen forces acting between and caused by relative movement between the device and the apparatus.

An apparatus for measuring flexion and extension of a joint of a user, said apparatus comprising:

- a first attachment and a second attachment, each comprising:
- a first layer with a top and a bottom, wherein the bottom comprises an adhesive material for coupling, on opposite sides of the joint, the attachments to the user;
- a second layer concentric with the first layer, and the second layer has a lower surface that couples to the first layer and an upper surface; and
- a pod concentric with the second layer and detachably coupled to the upper surface; and
- a goniometer for measuring flexion and extension of the joint of the user, and the goniometer is detachably coupled to, and spans between, the pods of the attachments; and
- a pedometer configured to count steps of the user and configured to be removably attached to the apparatus.

The apparatus wherein, to prevent the attachments from uncoupling from the user, the second layer comprises a material configured to dampen forces acting between the goniometer and the attachments and caused by relative movement between the goniometer and the attachments.

The apparatus wherein the goniometer has a pair of arms rotatably coupled to and about a center hub, and each arm has a boss for aligning and coupling each arm to a respective pod.

The apparatus wherein each pod defines a recess for receiving and detachably coupling to one of the bosses of the goniometer, and for positioning the goniometer relative to each attachment.

The apparatus wherein, to prevent the goniometer from uncoupling from the attachments, the bosses are sized to be movable within the recesses to allow movement of the bosses in the recesses.

The apparatus wherein, to detachably couple the goniometer to the attachments, magnets are disposed adjacent to each boss and each recess.

The apparatus wherein wings extend outwardly from the sides of each arm of the goniometer configured to enable a user to move the arms of the goniometer perpendicularly relative to the attachments, and to uncouple the goniometer from the attachments.

The apparatus wherein the edges of the pod are tapered, such configuration enabling a user to further move the arms of the goniometer perpendicularly relative to the attachments, and to uncouple the goniometer from the attachments.

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An apparatus for measuring flexion and extension at a joint of a user, the apparatus comprising:

a center hub having a first hub, and a second hub coaxially aligned with and rotatably coupled to the first hub;

a first arm coupled to and rotatable with the first hub; and

a second arm coupled to and rotatable with the second hub wherein, to maintain a position of the center hub relative to the joint of the user, the first and second arms pivotably and rotatably couple to a respective first and second hub such that rotation, flexion, and extension of the first and second arms are allowed.

The apparatus of claim **1**, wherein each of the first and second arms comprise an inner link for pivotably coupling the first and second arms to the first and second hubs, and the inner links are configured to allow the first and second arms to flex and extend relative to the center hub.

The apparatus wherein each of the first and second arms comprise an outer link coupled to the inner link, and the outer links are configured to rotate the first and second arms relative to the center hub.

The apparatus further comprising a screw configured to couple one of the inner and one of the outer links, wherein the screw is aligned parallel to a length of a respective one of the first and second arms, and wherein, to allow the rotation of the first and second arms relative to the center hub, the screw provides for rotation of the inner and outer links.

The apparatus wherein each of the first and second arms comprises an outer end coupled to the outer link, and the outer end is configured to flex and extend relative to the center hub such that further flexion and extension of the first and second arms relative to the center hub are enabled.

The apparatus further comprising link arms extending outwardly from each of the first and second hubs, wherein the link arms couple to respective ones of the first and second arms.

The apparatus further comprising pins configured to couple between a respective outer link and a respective outer end, and between a respective inner link and a respective link arm, and wherein the pins are aligned perpendicular to a length of a respective arm, and wherein the pins provide for the flexion and extension of the first and second arms relative to the center hub.

The apparatus wherein the arms include a magnet and a boss.

An apparatus for measuring flexion and extension at a joint of a user, the apparatus comprising:

a center hub that has an upper hub, and a lower hub coaxially aligned with and rotatably coupled to the upper hub;

an arm coupled to, and rotatable with, each of the upper and lower hubs, and the arms pivotably and rotatably couple to the hubs to allow rotation, flexion, and extension of the arms, and to restrict movement of the center hub relative to the joint of the user.

The apparatus further comprising link arms that extend outwardly from each of the upper and lower hubs, and the arms couple to a respective hub at a respective link arm.

The apparatus wherein pins couple each of the link arms and each of the arms, and the pins are aligned perpendicular to the length of a respective arm and configured to allow flexion and extension of the arms, and a screw couples between each of the link arms and each of the arms, and each screw is aligned parallel with the length of a respective arm configured to allow rotation of each arm about each screw.

The apparatus wherein the arms include a magnet and a boss for aligning and coupling the arms to attachments.

An apparatus for measuring flexion and extension at a joint of a user, the apparatus comprising:

a center hub comprising:

an upper hub and a first link arm that extends outwardly from the upper hub;

a lower hub coaxially aligned with and rotatably coupled to the upper hub, and a second link arm extends outwardly from the lower hub;

an arm pivotably coupled to each of the first and second link arms, and each arm comprises:

an inner link for pivotably coupling the arm to a respective first and second link arm and for allowing the arm to flex and extend relative to the center hub;

an outer link rotatably coupled to the inner link to allow the arm to rotate relative to the center hub;

an outer end pivotably coupled to the outer link to allow the outer end of the arm to flex and extend relative to the outer link and the center hub.

The apparatus wherein the link arms are integrally formed with the hubs.

The apparatus wherein the link arms are coupled to the respective hubs, and the link arms are configured to allow flexion and extension of the arms relative to the center hub.

The apparatus wherein the link arms couple to the respective hubs and are configured to allow rotation of the arms relative to the center hub.

The apparatus wherein a pin couples the inner link and the link arm, and the pin is aligned perpendicular to the length of the arm, and the pin provides for the pivotable coupling between the inner links and link arm.

The apparatus wherein a pin couples the outer link and the outer end, and the pin is aligned perpendicular to the length of the arm, and the pin provides for the pivotable coupling between the outer link and the outer end.

The apparatus wherein a screw couples the inner and outer link, and the screw is aligned parallel with the length of the arm, and the screw provides for the rotatable coupling between the inner link and outer link.

The apparatus wherein the arms include a magnet and a boss.

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An apparatus for assisting a person in aligning, relative to a center of a joint of a user, a wearable device on opposing limb portions of the joint of the user, the wearable device has attachments, each attachment defines alignment holes spaced from each other, and the alignment holes have a size, the apparatus comprising:

a first segment;

a second segment coupled to the first segment at a pivot point, and when positioned adjacent the center of the joint, the pivot point aligns the apparatus relative to the center of the joint;

the first and second segments each defining voids, the voids are spaced on each of the first and second segments equidistant from the pivot point, and the spacing between, and size of, the voids is commensurate with the spacing between, and size of, the alignment holes of the respective attachments; and

whereby the person pivots the segments to centrally align the segments relative to the respective opposing limb portions of the user, and the person marks locations on the respective opposing limb portions through the voids, wherein the alignment holes of the attachments are aligned with respective marks to align the attachments and to couple the wearable device to the user.

The apparatus wherein the first and second segments each have center axes, the center axes intersect at the pivot point, and prior to marking locations on the respective opposing limb portions, the person pivots each segment to centrally align the center axes with the respective centers of the opposing limb portions.

The apparatus wherein the second segment pivotably couples to the first segment at the pivot point.

The apparatus wherein the second segment couples to the first coupling end at a pivot point by overlaying with the first segment.

The apparatus wherein the pivot point is spaced adjacent to and equidistant from first and second coupling ends of the respective first and second segments.

The apparatus wherein the alignment holes of the attachments are aligned coaxially with respective marks to align the attachments, and enable the wearable device to be mounted on the user.

A method for assisting a person in aligning a wearable device relative to a center of a joint of a user, on opposing limb portions of the joint of the user, the wearable device has attachments, each attachment defines alignment holes spaced from each other, and each alignment hole has a size, the method comprising:

providing an apparatus having a first segment and a second segment pivotably coupled to the first segment at a pivot point;

providing voids defined in each of the first and second segments, the voids are spaced an equidistant from the pivot point on each of the first and second segments, and the spacing between, and size of, the voids is commensurate with the spacing between, and size of, the alignment holes of the respective attachments;

locating the center of the joint;

aligning the pivot point of the apparatus with the center of the joint;

pivoting each segment to centrally align the segments relative to respective opposing limb portions;

marking locations on the respective opposing limb portions through the voids;

positioning the alignment holes of the attachments with respective marks to align the attachments and the wearable device on the user; and

coupling the attachments and the wearable device to the user.

The method further comprising:

providing center axes of the first and second segments, and the center axes intersect at the pivot point;

pivoting each segment to centrally align the center axes with the center of the respective opposing limb portion.

The method further comprising:

providing pegs; and

coaxially aligning the pegs with the markings to facilitate alignment of the alignment holes of the attachments with each mark.

The method wherein the step of locating the center of the joint is further defined as locating the lateral epicondyle of a knee joint.

An apparatus for assisting a person in aligning a wearable device relative to a center of a joint of a user, on opposing limb portions of the joint of the user, the wearable device has attachments, each attachment defines alignment holes spaced from each other, and the alignment holes have a size, the apparatus comprising:

an imaging device for locating a position of the joint of the user and the opposing limb portions;

a laser configured to provide a beam of light;

a processor in communication with the imaging device and the laser; and

the processor is configured to receive and process data from the imaging device representative of the position of the joint of the user and the opposing limb portions, the processor communicates with the lasers to position the laser such that the light beam forms spots of light on the user at the center of the joint and on the opposing limb portions of the user commensurate with the spacing between, and size of, the alignment holes of the respective attachments, and wherein the alignment holes of the attachments are aligned with respective alignment holes of the attachments to couple the wearable device on the user.

A method for assisting a person in aligning a wearable device relative to a center of a joint of a user, on opposing limb portions of the joint of the user, the wearable device has attachments, each attachment defines alignment holes spaced from each other, and each alignment hole has a size, the method comprising:

providing an imaging device;

providing a laser configured to provide a beam of light;

providing a processor configured to communicate with the imaging device and the laser;

locating with the imaging device positions of the joint and the opposing limb portions;

communicating data from the imaging device to the processor, the data being representative of the positions of the joint and the opposing limb portions;

processing the data with the processor;

calculating with the processor positions of the laser to provide the laser beam to form spots of light on the user at the center of the joint and the opposing limb portions, and the foregoing is commensurate with the spacing between, and size of, the alignment holes of the respective attachments;

communicating instructions to the laser;

moving the laser to the positions; and

aligning the alignment holes of the attachments with the respective spots of light to align the attachments and to couple the wearable device to the user.

The method further comprising the step of marking on the opposing limb portions the location of the laser beam, and aligning the alignment holes of the attachments coaxial with respective marks to align the attachments and to couple the wearable device to the user.

An apparatus for assisting a person in aligning a wearable device relative to a center of a joint of a user and on opposing limb portions of the joint of the user, the apparatus comprising:

diodes coupled to the wearable device, the joint, and the opposing limb portions;

an imaging device for locating positions of the diodes;

a personal communication device configured to communicate with a person;

a processor in communication with the imaging device and the personal communication device; and

the processor is configured to receive and process data from the imaging device representative of the positions of the diodes, and the processor communicates with the personal communication device to provide instructions representative of action to be taken by the person to align the wearable device relative to the center of joint of the user.

The apparatus wherein the personal communication device is a display.

The apparatus wherein the personal communication device is a speaker.

A method for assisting a person in aligning a wearable device on opposing limb portions of a joint of a user relative to a center of the joint of the user, the method comprising:

providing a diode coupled to the wearable device, the joint, and the opposing limb portions of the user;

providing an imaging device for locating a position of the diodes;

providing a personal communication device configured to communicate with a person;

providing a processor configured to communicate with the imaging device and the personal communication device;

locating with the imaging device the position of the diodes;

communicating, from the imaging device to the processor, data representative of the position of the diodes;

processing of the data by the processor;

calculating by the processor a position of the wearable device relative to the joint and the opposing limb portions;

calculating by the processors a position of the wearable device, and instructions for the person to align the wearable device relative to the center of the joint of the patient;

communicating by the processors to the personal communication device the instructions to be provided to the person for aligning the wearable device relative to the center of the joint of the user; and

communicating to the person with the personal communication device.

The method wherein the step of communicating to the person with the personal communication device further comprises providing audio instruction through a speaker to the personal communication device.

The method wherein the step of communication to the person with the communication device further comprises providing visual instruction through a display to the personal communication device.

An apparatus for assisting a person in aligning, relative to a center of a joint of a user, a wearable device on opposing limb portions of the joint of the user, the wearable device has attachments, each attachment defines alignment holes spaced from each other, and the alignment holes have a size, the apparatus comprising:

a first segment;

a second segment coupled to the first segment at a pivot point, and when positioned adjacent the center of the joint, the pivot point aligns the apparatus relative to the center of the joint;

the first and second segments each defining voids, the voids are spaced on each of the first and second segments equidistant from the pivot point, and the spacing between, and size of, the voids is commensurate with the spacing between, and size of, the alignment holes of the respective attachments; and

whereby the person pivots the segments to centrally align the segments relative to the respective opposing limb portions of the user, and the person marks locations on the respective opposing limb portions through the voids, wherein the alignment holes of the attachments are aligned with respective marks to align the attachments and to couple the wearable device to the user.

The apparatus wherein the first and second segments each have center axes, the center axes intersect at the pivot point, and prior to marking locations on the respective opposing limb portions, the person pivots each segment to centrally align the center axes with the respective centers of the opposing limb portions.

The apparatus wherein the second segment pivotably couples to the first segment at the pivot point.

The apparatus wherein the second segment couples to the first coupling end at a pivot point by overlaying with the first segment.

The apparatus wherein the pivot point is spaced adjacent to and equidistant from first and second coupling ends of the respective first and second segments.

The apparatus wherein the alignment holes of the attachments are aligned coaxially with respective marks to align the attachments, and enable the wearable device to be mounted on the user.

A method for assisting a person in aligning a wearable device relative to a center of a joint of a user, on opposing limb portions of the joint of the user, the wearable device has attachments, each attachment defines alignment holes spaced from each other, and each alignment hole has a size, the method comprising:

providing an apparatus having a first segment and a second segment pivotably coupled to the first segment at a pivot point;

providing voids defined in each of the first and second segments, the voids are spaced an equidistant from the pivot point on each of the first and second segments, and the spacing between, and size of, the voids is commensurate with the spacing between, and size of, the alignment holes of the respective attachments;

locating the center of the joint;

aligning the pivot point of the apparatus with the center of the joint;

pivoting each segment to centrally align the segments relative to respective opposing limb portions;

marking locations on the respective opposing limb portions through the voids;

positioning the alignment holes of the attachments with respective marks to align the attachments and the wearable device on the user; and

coupling the attachments and the wearable device to the user.

The method further comprising:

providing center axes of the first and second segments, and the center axes intersect at the pivot point;

pivoting each segment to centrally align the center axes with the center of the respective opposing limb portion.

The method further comprising:

providing pegs; and

coaxially aligning the pegs with the markings to facilitate alignment of the alignment holes of the attachments with each mark.

The method wherein the step of locating the center of the joint is further defined as locating the lateral epicondyle of a knee joint.

An apparatus for assisting a person in aligning a wearable device relative to a center of a joint of a user, on opposing limb portions of the joint of the user, the wearable device has attachments, each attachment defines alignment holes spaced from each other, and the alignment holes have a size, the apparatus comprising:

an imaging device for locating a position of the joint of the user and the opposing limb portions;

a laser configured to provide a beam of light;

a processor in communication with the imaging device and the laser; and

the processor is configured to receive and process data from the imaging device representative of the position of the joint of the user and the opposing limb portions, the processor communicates with the lasers to position the laser such that the light beam forms spots of light on the user at the center of the joint and on the opposing limb portions of the user commensurate with the spacing between, and size of, the alignment holes of the respective attachments, and wherein the alignment holes of the attachments are aligned with respective alignment holes of the attachments to couple the wearable device on the user.

A method for assisting a person in aligning a wearable device relative to a center of a joint of a user, on opposing limb portions of the joint of the user, the wearable device has attachments, each attachment defines alignment holes spaced from each other, and each alignment hole has a size, the method comprising:

providing an imaging device;

providing a laser configured to provide a beam of light;

providing a processor configured to communicate with the imaging device and the laser;

locating with the imaging device positions of the joint and the opposing limb portions;

communicating data from the imaging device to the processor, the data being representative of the positions of the joint and the opposing limb portions;

processing the data with the processor;

calculating with the processor positions of the laser to provide the laser beam to form spots of light on the user at the center of the joint and the opposing limb portions, and the foregoing is commensurate with the spacing between, and size of, the alignment holes of the respective attachments;

communicating instructions to the laser;

moving the laser to the positions; and

aligning the alignment holes of the attachments with the respective spots of light to align the attachments and to couple the wearable device to the user.

The method further comprising the step of marking on the opposing limb portions the location of the laser beam, and aligning the alignment holes of the attachments coaxial with respective marks to align the attachments and to couple the wearable device to the user.

An apparatus for assisting a person in aligning a wearable device relative to a center of a joint of a user and on opposing limb portions of the joint of the user, the apparatus comprising:

diodes coupled to the wearable device, the joint, and the opposing limb portions;

an imaging device for locating positions of the diodes;

a personal communication device configured to communicate with a person;

a processor in communication with the imaging device and the personal communication device; and

the processor is configured to receive and process data from the imaging device representative of the positions of the diodes, and the processor communicates with the personal communication device to provide instructions representative of action to be taken by the person to align the wearable device relative to the center of joint of the user.

The apparatus wherein the personal communication device is a display.

The apparatus wherein the personal communication device is a speaker.

A method for assisting a person in aligning a wearable device on opposing limb portions of a joint of a user relative to a center of the joint of the user, the method comprising:

providing a diode coupled to the wearable device, the joint, and the opposing limb portions of the user;

providing an imaging device for locating a position of the diodes;

providing a personal communication device configured to communicate with a person;

providing a processor configured to communicate with the imaging device and the personal communication device;

locating with the imaging device the position of the diodes;

communicating, from the imaging device to the processor, data representative of the position of the diodes;

processing of the data by the processor;

calculating by the processor a position of the wearable device relative to the joint and the opposing limb portions;

calculating by the processors a position of the wearable device, and instructions for the person to align the wearable device relative to the center of the joint of the patient;

communicating by the processors to the personal communication device the instructions to be provided to the person for aligning the wearable device relative to the center of the joint of the user; and

communicating to the person with the personal communication device.

The method wherein the step of communicating to the person with the personal communication device further comprises providing audio instruction through a speaker to the personal communication device.

The method wherein the step of communication to the person with the communication device further comprises providing visual instruction through a display to the personal communication device.

The above discussion is meant to be illustrative of the principles and various embodiments of the present invention. Numerous variations and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

The various aspects, embodiments, implementations or features of the described embodiments may be used separately or in any combination. The embodiments disclosed herein are modular in nature and may be used in conjunction with or coupled to other embodiments.

Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any feature(s) that can cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, sacrosanct or an essential feature of any or all of the claims.

Consistent with the above disclosure, the examples of assemblies enumerated in the following clauses are specifically contemplated and are intended as a non-limiting set of examples.

What is claimed:

1. A system for measuring an angle of a joint of a user, comprising:
- a center hub, the center hub having a first hub and a second hub;
- a first coupling apparatus and a second coupling apparatus external to the center hub, each of the first coupling apparatus and the second coupling apparatus having a first layer and a second layer disposed above the first layer, wherein the first layer defines an area larger than an area of the second layer, and wherein the first and second coupling apparatuses are configured to be adhesively attached to first and second limb portions proximal to the joint of the user, respectively;
- a first arm including a first outer end and a first inner end opposite the first outer end, wherein the first arm is attached to the second layer of the first coupling apparatus at the first outer end and the first arm is further attached to the first hub at the first inner end;
- a second arm including a second outer end and a second inner end opposite the second outer end, wherein the second arm is attached to the second layer of the second coupling apparatus at the second outer end and the second arm is further attached to the second hub at the second inner end, and, further, wherein the first hub is pivotally coupled to the second hub;
- a magnet coupled to the second hub;
- a sensor configured to detect a rotation of the magnet; and
- a transmitter configured to transmit an electrical signal based on the rotation of the magnet, wherein the electrical signal has data of the angle of the joint of the user.

2. The system of claim 1, further comprising a cover coupled to the center hub, wherein the cover has a snap mechanism configured to attach and detach the cover.

3. The system of claim 2, wherein the cover is detachably coupled to the first hub.

4. The system of claim 1, wherein the first and second arms have sections configured to rotate the first arm about the first inner end and the second arm about the second inner end each about +/−10 degrees.

5. The system of claim 1, wherein the first and second arms have sections configured to twist the first arm about the first inner end and the second arm about the second inner end each about +/−18 degrees.

6. The system of claim 1, further comprising a retaining ring configured to couple the magnet to the second hub.

7. The system of claim 6, wherein the retaining ring is coupled to a magnet housing, wherein the retaining ring is configured to secure the magnet within the magnet housing.

8. The system of claim 1, wherein sensor is configured to detect the rotation of the magnet at a sensitivity of a one-hundredth of a degree.

9. The system of claim 1, wherein the first layer defines at least one notch in a peripheral edge thereof for marking a skin there beneath and to facilitate aligning a replacement coupling apparatus.

10. The system of claim 9, wherein the at least one notch has a V-shape.

11. The system of claim 9, wherein the at least one notch includes at least two notches spaced apart from one another.

12. The system of claim 9, wherein the at least one notch includes four notches each spaced apart from one another.

13. The system of claim 1, further comprising a pod attached one of the second layers, opposite from the first layer, and wherein the pod is configured to be releasably attached to one of the first arm or the second arm.

14. The system of claim 13, wherein the pod is secured to the one of the first arm or the second arm with a magnet.

15. The system of claim 13, wherein at least one of the pod, the first arm, or the second arm defines one or more recesses configured to receive a boss and to hold the pod in a fixed position when the pod is attached to the one of the first arm or the second arm.

16. A system for measuring an angle of a joint of a user, comprising:
- a center hub, the center hub having a first hub and a second hub;
- a first coupling apparatus and a second coupling apparatus external to the center hub, each of the first coupling apparatus and the second coupling apparatus having a layer defining at least one notch, wherein the first coupling apparatus and second coupling apparatus are configured to be adhesively attached to first and second limb portions proximal to the joint of the user, respectively;
- a first arm including a first outer end and a first inner end opposite the first outer end, wherein the first arm is attached the first coupling apparatus at the first outer end and the first arm is further attached to the first hub at the first inner end;
- a second arm including a second outer end and a second inner end opposite the second outer end, wherein the second arm is attached to the second coupling apparatus at the second outer end and the second arm is further attached to the second hub at the second inner end, and, further, wherein the first hub is pivotally coupled to the second hub; and
- a sensor configured to detect a rotation of the second hub relative to the first hub.

17. The system of claim 16, wherein the at least one notch has a V-shape.

18. The system of claim 16, wherein the at least one notch includes at least two notches spaced apart from one another.

19. The system of claim 16 further comprising a pod attached one of the first coupling apparatus or the second coupling apparatus, and wherein the pod is configured to be releasably attached to one of the first arm or the second arm.

20. The system of claim 19, wherein the pod is secured to the one of the first arm or the second arm with a magnet.

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